GAIT BASED HUMAN MOVEMENT OBSERVATION USING SILHOUETTE

Zaheer Hussain Fida

(FA21-RCS-005)

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Department of Computer Science and Information Technology

Faculty of Natural and Applied Science

Mirpur University of Science and Technology (MUST), Mirpur

(AJK), Pakistan.

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By

Zaheer Hussain Fida

(MUST/FA21-RCS-005/AJK)

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(AJK), Pakistan.

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Registration No: **FA21-RCS-005** Date: _____

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Supervisor: _____

Co-Supervisor: _____

External Examiner: _____

Chairman: _____

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Director Advance Studies & Research Board: _____

DEDICATION

This thesis is dedicated to my beloved parents, my supervisor Dr. Yasir Mahmood and co- supervisor Dr. Sajid Hussain. To my parents, your unwavering belief and provision of resources, have nurtured my intellect and helped me explore new courses. To my exceptional supervisor, Dr. Yasir Mahmood, a beacon of knowledge, your expertise and guidance have been my compass and leverage, but it is through much guidance and motivation of my supervisor. Dr. Sajid Hussain was a source of unwavering motivation, support, and countless resources provided to me during the research.

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ABBREVIATIONS

ACL: Anterior Cruciate Ligament

AFDEI: Accumulated Frame Difference Energy Image

AI: Artificial Intelligence

BGRU: Bidirectional Gated Recurrent Unit

CASIA: Chinese Academy of Sciences Institute of Automation

CCA: Canonical Correlation Analysis

CGAN: Conditional Generative Adversarial Network

CNN: Convolutional Neural Network

D: Dimension

DTW: Dynamic Time Warping

EMG: Electromyography

FC: foot-contact

FO: foot-off

GGEI: Gradient Gait Energy Image

GUI: Graphical User Interface

HT: Hough Transform

IMU: Inertial Measurement Unit

INIT: Institute of New Imaging Technologies

ISIR: Institute of Scientific and Industrial Research

KGB: Kinect-Based Gait

LABCOM: Laboratory of Communication Sciences

LFN: Leaky Fusion Neuron

LU: Leaky ReLU (Rectified Linear Unit)

LSTM: Long Short-Term Memory

LP: Large Population

MGAN: Multi-Generator Adversarial Network

ML: Machine Learning

NN: Neural Network

OP: Open Pose

OU-ISIR: Osaka University - Institute of Scientific and Industrial Research

PD: Parkinson's disease

PEI: Period Energy Image

PSTN: Pairwise Spatial Transformer Network

RB: Recurrent Neural Network

RGB: Red Green Blue

RMS: Root Mean Squared

ROC: Receiver Operating Characteristic

SG: Spectrogram

SR: Set Residual

SRN: Set Residual Network

SVM: Support Vector Machine

TGLSTM: Triplet Gate Long Short-Term Memory

Wix: Weight matrix associated with input features

Wih: Weight matrix associated with previous hidden state

Wfx: Weight matrix associated with input features

Wfh: Weight matrix associated with previous hidden state

Wcx: Weight matrix associated with input features

Wch: Weight matrix associated with previous hidden state

Wox: Weight matrix associated with input features

Woh: Weight matrix associated with previous hidden state

ABSTRACT

Human movement describes how people move and it plays a vital role in many social and cultural implications. This movement is based on physical actions of muscles, joints, and nervous system. In many fields of daily life like sports, health, military recruitment, etc., we need to observe how humans move while walking and running. Gait analysis is a structured examination of how a person walks or moves, which includes measuring and appraising a range of biomechanical and kinematic factors to assess their movement patterns. This walking pattern involves trained observers to examine various aspects of movement. Gait movement patterns provide valuable information for clinicians and physical therapists in diagnosing gait disorders like Parkinson's disease and guiding treatment planning. Gait analysis methods include gait identification and gait verification. The gait identification distinguishes individual by analyzing unique walking pattern, aiding biometric authentication, surveillance, and forensics while the gait recognition matches gait patterns to a database without revealing personal information, often used for security and access control. The gait recognition is based on multi-faceted approach encompass sensor-based, sound-based, and frame-based. The sensor-based approach involved leveraging sensors to capture and interpret intricate movement of patterns. The sound-based technique is employed to enhance the understanding of motion sequences. The frame-based method is used to dissect and analyze the postures and alignment of individuals during activities such as walking and running. The research focuses on developing a system for monitoring and evaluating the fitness levels of military personnel during recruitment and throughout their services. A transfer learning frame-based gait recognition approaches AlexNet and AlexLSTM are implemented to identify whether the movement is normal or abnormal. The experiments are performed on silhouette based OUISIR, OUISIR ST, and INIT gait datasets and achieved accuracy of AlexNet 98.69%, 99.40%, 99.71% and AlexLSTM 99.14%, 99.81%, 99.96% respectively. The proposed method has wide-ranging implications for both medical and military applications contributing to improved gait recognition.

Chapter No. 01

INTRODUCTION

Human movement is a multifaceted concept that encompasses a wide range of physical activities and actions integral to our daily lives. It plays a fundamental role in our overall well-being, promoting both physical and mental health through regular exercise and physical activity. It involves the development of specific skills, sports, health, military recruitment, serving as a source of personal achievement. Human movement is the visual and qualitative assessment of an individual's walking pattern. It involves for observers to visually examine various aspects of a person’s moves while walking. Human movement describes how people walk and it plays a vital role in many social and cultural implications. In socially, it is fosters interaction and community building, with team sports and group fitness classes offering opportunities to connect with others. In cultural, it is encompass dance, martial arts, rituals, and nonverbal cues, reflecting and preserving the values, traditions, and identity of a given culture. In socially and culture environments, the human movement observation plays important role during the walking [1].

Human movement observation is the systematic and structured process of watching and analyzing the physical actions, gestures, and behaviors of individuals. The primary objective of human movement observation is to gather accurate and valuable information about how people move. This information is then used to achieve specific goals, whether they are related to sports, health and military recruitment for clinical and system environments[1].

In clinical and system environments, human movement observation plays a critical role. It impacts clinical practice to make treatment and helps therapists design tailored interventions to improve gait patterns in patients. The therapies are improving walking patterns for individuals with musculoskeletal disorders, neurological impairments, and injuries affecting gait. In system environments, hospitals, or remote healthcare settings where advanced technologies are limited and human movement observation remains a crucial assessment tool. The assessment and monitoring of patients' movements and overall mobility are crucial. Its simplicity ensured quick initial evaluations and allowed for timely care for individuals with walking related issues like Parkinson’s disease.

Parkinson’s disease is an advancing neurological based condition characterized by the gradual decline in motor skills that control the person activity. It is affected individuals to

face unique challenges of walking during their service. It may lead to medical evaluations to assess the impact on their ability to fulfill their service. Depending on the severity and progression of the disease, it is access to healthcare, disability benefits, and pre and post-service support for gait analysis of individuals[2].

Gait analysis is a method used to analyze an individual's walk pattern by capturing and studying the series of data. This method relies on high-speed that record individual's movements as they walk along a designated path or specialized walkway. Gait analysis is key aspects of individual's walking pattern are assessed include joint angles, limb movements, body posture, and foot placement [3]. When conducting gait analysis, a series of key parameters are typically measured and assessed. The parameters are stride length, step duration, and joint angles. These parameters yield invaluable insights into an individual's gait, enabling the identification of abnormalities and areas for improvement. This systematic evaluation of gait plays a crucial role in fields like healthcare, sports and military recruitment. Gait analysis can be employed for gait identification and recognition as a means of enhancing security and surveillance [4].

The existing methods are gait identification and gait recognition. Gait identification involves distinguishing and recognizing individuals by analyzing their distinct walking patterns This system is focused on analyzing the characteristics of individual's gait pattrens to establish their identification. Gait identification systems are useful in applications such as biometric authentication, surveillance, or forensic investigations. Gait recognition entails identifying individuals based on their unique walking patterns without the need to establish their specific names or personal information. The primary purpose of this system is to classify and match gait patterns to a pre-existing database of gait templates for military personnel’s. Gait identification and recognition rely on various approaches for analyzing gait pattern [14].

A silhouette refers to a two-dimensional outline or shadow-like representation (black and white) of an individual's body. Individual's body while they are in motion, typically walking or running. Silhouettes are created by capturing images or video frames of the person from a specific angle. Angles are often from the side, and then traced or extracted the outer contour of the body in each frame. Silhouettes are used in walk analysis to the general body posture and movement patterns during walking or running. The tracking is important for silhouette over time and assessed the change in body shape, position from one frame to another while

walking of personnel. The method is used by analysts to derive the information for different aspects of walking [4].

During the walking, the different information is analysis like stride length, step duration, limb movement, and posture from silhouette. The silhouette is contains information in which it categorized into normal and abnormal movement from the walking of individuals. It contains the video of frame and these videos for each individual contain the specific speed of frames during the walking. Silhouette is particularly useful when more sophisticated motion capture systems or high-speed cameras are not available. Silhouette is employed to observe and evaluate movement patterns. Silhouette analysis involves extracting and analysing the shape and motion information of individuals from silhouette images or video sequences. This non-intrusive method allows for the assessment of walking patterns of individual and the identification of potential abnormalities or deviations from walking. These individual contains silhouette based normal walking on wall alignment of head, arm swing and foot stride if these are not on well form these are abnormal walking shown in figure 1.1.

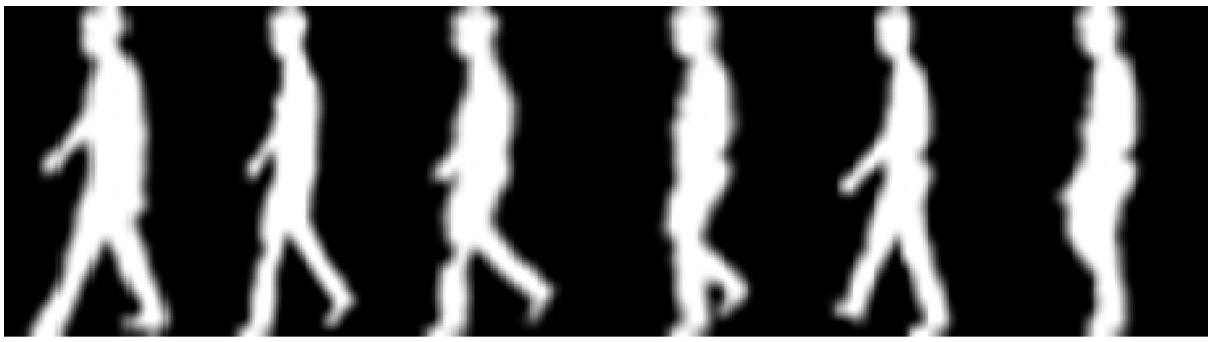


Figure 1.1: Walking pattern/Silhouette

Silhouette-based walk analysis provides an objective and systematic approach to assessed soldiers' walking styles. By leveraging silhouette data and analysed gait parameters, military trainers and medical professionals make informed decisions regarded training, intervention, and task assignments. The ultimate aim is to optimized soldiers' gait patterns, ensured their effectiveness, safety, and overall well-being in military operations during the training [6].

For military personnel, training included drills and marching techniques that aim to in still discipline, precision, and uniformity among soldiers. One of the key aspects of military drill is the way soldiers walk in formations, which differs from normal civilian walking. This distinctive walking style is essential for soldiers to maintain proper alignment and behave like a normal person. Soldiers focus on their posture during marching and maintain an upright position with their heads aligned vertically, chins up, and eyes fixed straight ahead. This

posture ensured a command presence and reflected the professionalism expected in the military [2].

In addition to posture, soldiers pay close attention to their arm swing. The swinging of arms is synchronized with their steps and creates a harmonious movement pattern. The soldiers take a stride with their right foot, the left arm swings forward and mutually. This deliberate arm swing contributed to the overall sense of uniformity and precision. Another crucial aspect is stride. Soldiers take measured strides, ensured that each step covers a consistent distance. These strides are often longer than those in normal walking, allowed soldiers to cover more ground efficiently. The uniformity of stride among soldiers is vital for maintain synchronized alignment and prevented disruptions in movement [3] as shown in figure 1.2 which is collected from Google search engine.

Figure 1.2: Military Personnel's Walk

Maintaining alignment is also of most important. Soldiers strive to walk in straight lines with equal space between individuals in the formation. Their shoulders, hips, and feet align with those of their fellow soldiers, creates a visually strike display of discipline and order [2]. These manoeuvres require coordination and practice to execute flawlessly. The ability to turn and pivot smoothly ensured that soldiers change direction and maintain formation integrity [3].

There are multifaceted approaches like sound-based, sensor-based and frame-based for gait recognition. Sound-based analysis uses audio sensors to capture footstep sounds during walking, making it suitable for gait analysis [21]. Sensor-based analysis involves various sensors like IMUs to gather data on joint angles, muscle activation, and ground forces for gait analysis. It is versatile for clinical, rehabilitation, sports, and research applications. These techniques collectively aid in understand and assessed human walking patterns. It is supported in clinical diagnosis, treatment planning, biomechanical research, and performance optimization across diverse fields for gait analysis [19]. Frame-based analysis utilizes high-speed cameras to record detailed movements. It provided comprehensive insights into joint

angles and gait mechanics, valuable for biomechanical research and clinical assessments [52]. The selection of these approaches is to rely on particular objectives and evaluation for gait analysis to detect the normality or abnormality for military personals.

The evaluation of human walking actions plays a pivotal role in military. It serves to differentiate between normal and abnormal movement patterns for military personals. This distinction is particularly significant in the context of military personnel, for whom physical fitness and dynamic stability are of utmost importance. Military personnel are frequently engaged in strenuous exercises and activities that involve significant overuse of musculoskeletal demands. Surprisingly, there currently exists no comprehensive system capable of determining an individual's suitability for military service in terms of physical fitness. Consequently, it becomes imperative to assess the fitness and dynamic stability of military recruits based on gait analysis, both during the recruitment process and throughout their service. This research is not only aids in identifying potential issues but also helps in tailoring training and interventions to ensure the well-being and performance of military personnel in physically demanding roles using deep learning [1].

In Deep learning, AlexNet is CNN architecture, acclaimed for its performance in image classification tasks. LSTM is RNN architecture effective in handling sequential data with long-range dependencies. Sequential data are time series and natural language data. Both are influential components of deep learning, with AlexNet excelled in deep learning applications. LSTM is prominent in natural language processing and sequential data analysis in gait analysis and combine with pre-train model AlexNet using transfer learning [42].

Transfer learning is a powerful technique that enhanced the performance of a model on a new, related task by leveraging knowledge from a previously trained task. This method is starting with a pre-trained model, typically it is train on a vast and diverse dataset which are already learned valuable features and patterns. This existing knowledge proved invaluable when dealing with limited data for the new task. Transfer learning is executed through two common approaches. One is feature extraction, where the pre-trained model's fixed features were used and second is fine-tuning, which involved adjusting the pre-trained model's weights to tailor it to the new task's requirements. Overall, transfer learning is a critical tool in the machine learning toolbox for efficiently applying deep learning models to various tasks. The dataset are based on silhouette and it is special form in to train with different techniques of transfer learning for our research.

1.1 PROBLEM STATEMENT

The evaluation of human walking action is applied in various fields to distinguish between normal and abnormal movement. For military personnel’s physical fitness and dynamic stability is extremely important as military personals are exposed to extensive exercises and activity involving musculoskeletal overuse. There is no existing system capable of determined an individual's physical fitness for military service. Therefore, it is crucial to assess the military personnel’s fitness and dynamic stability based on gait analysis at the time of recruitment as well as during their service. In the realm of assessing human movement, particularly in the context of military personnel, the evaluation of walking actions serves as a critical tool to discern between normal and abnormal gait patterns.

Consequently, it becomes imperative to incorporate gait analysis as a pivotal component in assessing military personnel's fitness and dynamic stability, both at the time of recruitment and throughout their service. By leveraging sophisticated technologies and methodologies in gait analysis, such as motion capture systems and biomechanical modeling, it becomes possible to gain profound insights into the intricacies of an individual's walking patterns. This not only aids in identifying potential abnormalities but also allows for a nuanced evaluation of overall physical health and performance.

1.2 RESEARCH QUESTIONS

The research explored the multifaceted domain of gait analysis, investigating its critical role in understanding human movement patterns across diverse fields such as sports, health, and military recruitment. Biomechanical and kinematic factors are examined in this structured analysis to assess movement patterns, particularly aiding clinicians in diagnosing gait disorders like Parkinson's disease. The study delves into the application of gait identification for biometric authentication, surveillance, and forensics, emphasizing the unique walking patterns of individuals. Furthermore, the research scrutinizes sensor-based, sound-based, and frame-based approaches in gait recognition, examining their contributions to capturing intricate movement patterns and enhancing the understanding of motion sequences. A novel focus on frame-based gait recognition, specifically employing the AlexNet and AlexLSTM models, is proposed for monitoring and evaluating the fitness levels of military personnel during recruitment and service. Silhouette-based datasets, such as OUISIR, OUISIR ST, and INIT, are utilized for experimentation, demonstrating high accuracy in gait recognition. The study also considers ethical and privacy implications, anticipating the potential of this

research to advance both medical diagnosis and military applications through improved gait recognition technology, contributing to the broader understanding of human movement patterns are as follow.

Q1. How does gait analysis contribute to the understanding of human movement patterns in various fields, such as sports, health, and military recruitment?

Q2. What biomechanical and kinematic factors are commonly measured and appraised during gait analysis to assess movement patterns?

Q3. In what ways does gait analysis aid clinicians and physical therapists in diagnosing gait disorders, particularly in conditions like Parkinson's disease?

Q4. How can gait identification be effectively utilized for biometric authentication, surveillance, and forensics, considering the unique walking patterns of individuals?

Q5. What are the key components and features of sensor-based gait recognition, and how do these sensors capture and interpret intricate movement patterns?

Q6. How does the sound-based approach contribute to enhancing the understanding of motion sequences in gait recognition?

Q7. What role does the frame-based method play in dissecting and analyzing postures and alignment during activities such as walking and running in gait recognition?

Q8. How can gait recognition methods be applied to monitor and evaluate the fitness levels of military personnel during recruitment and throughout their services?

Q9. What challenges and limitations exist in the implementation of gait recognition systems for military applications, and how can they be addressed?

Q10. How does the transfer learning approach enhance the performance of gait recognition systems, specifically in the context of military personnel fitness monitoring?

Q11. What are the differences in performance and accuracy between the AlexNet and AlexLSTM models in the context of frame-based gait recognition?

Q12. How do silhouette-based datasets, such as OUISIR, OUISIR ST, and INIT, contribute to the evaluation and testing of gait recognition systems for military fitness monitoring?

Q13. What ethical considerations and privacy concerns arise when deploying gait recognition systems, especially in military and security applications?

Q14. How can the proposed gait recognition system be adapted or extended for real-time monitoring of military personnel's fitness during dynamic activities?

Q15. What are the potential implications and future directions of the research in terms of advancing both medical and military applications through improved gait recognition technology?

1.3 CONTRIBUATION

The research contribution aimed to develop a system specifically for military applications, focused on monitoring and evaluating the fitness levels of military personnel's during recruitment and throughout their services.

- The literature review revealed various pathways for conducted gait analysis, with sensor-based, sound-based, and frame-based methods emerged as the primary techniques employed to gain insights into human movement patterns.
- Models AlexNet, and AlexLSTM was used as deep learning for extraction of intricate patterns and features from data, and train on three dataset for movement assessment either the military personnel movement is normal or abnormal.

1.4 RESEARCH OBJECTIVE

The aim of this research is to create an effective ambulation assessment system that identify whether a person is physically normal or not with a particular emphasis on assessed physical fitness and dynamic stability in military personnel. By investigated and integrated advanced deeplearing methods, the aimed is to create a practical tool that accurately classified gait patterns as normal or abnormal movement. Aim to provide an effective means of evaluated and monitored the physical fitness and dynamic stability of military personnel. Physical fitness and dynamic stability of military personnel during recruitment and throughout their service, thereby enhanced their overall performance and reduced the risk of musculoskeletal injuries.

1.5 RESEARCH GAPE

Gait analysis stands as a critical element in the comprehensive evaluation of the fitness and performance of military personnel. Despite its significance, a noticeable gap exists in the absence of diverse gait datasets explicitly designed and utilized for military purposes. This dearth suggests that the military sector has yet to fully capitalize on the potential of gait analysis to bolster security measures and enhance operational effectiveness within its ranks. The deficiency becomes more apparent when considering the crucial role that gait identification and recognition play in military applications, such as personnel identification and movement detection. The absence of tailored datasets specifically for military personnel limits the accuracy and relevance of gait analysis in addressing the unique demands and requirements of the armed forces.

To bridge this research gap, there is a compelling need to leverage diverse gait datasets, including those related to INIT Gait, OUISIR ST, and OUISIR, which could provide invaluable insights for various military tasks. These datasets encompass tasks related to gait identification and recognition, which are pivotal for applications like personnel identification and movement detection in military contexts. Incorporating such diverse datasets tailored to the military domain not only facilitates a more accurate understanding of the intricacies of military personnel's movement patterns but also opens avenues for the development of specialized gait analysis tools that can significantly contribute to the security and operational efficiency of military operations.

Chapter No. 02

BACKGROUND AND LITERATURE REVIEW

2.1 INTRODUCTION

Gait analysis, is a multidisciplinary field that encompassed various domains, systems, and methodologies. It spans across different domains for its applications, included clinical rehabilitation, sports performance optimization, and biomechanical. Gait analysis systems are employed advanced technologies to collect and process data [7]. Gait Analysis Systems plays a central role in the realms of Gait Identification and Gait Recognition/Verification. These advanced technologies enable precise assessment to understand of an individual's walking patterns. This offered valuable applications in various fields, from biometrics to healthcare. Gait analysis encompasses three methods frame-based, sensor-based, and sound-based. Frame-based analysis employs high-speed cameras to meticulously study an individual's movement that provided detailed insights into biomechanics. Sensor-based approaches use a range of sensors like IMUs, force plates, and pressure sensors to comprehensively assess gait patterns [8]. Sound-based gait analysis relies on audio sensors or microphones to capture and analyse walking sounds which offered an alternative especially in challenging scenarios [9]. These methods collectively enhanced understanding of human locomotion, guided personalized interventions and promoted overall well-being.

2.2 DOMAINS FOR GAIT ANALYSIS

There are different domains in which gait analysis plays vital role in medical field, which are disused below in detail.

2.2.1 Medical and Rehabilitation

Gait analysis plays a crucial role in medical and rehabilitation settings, where it is used to diagnose and monitor a range of conditions. These conditions are musculoskeletal disorders, Parkinson's disease, orthopaedic injuries, and prosthetic/orthotic fitting [7]. By objectively measured and analysed an individual's gait patterns, healthcare professionals gained valuable insights. It is also analysed for gait deviations, asymmetries, joint range, muscle imbalances, and compensatory movements. These insights aid in treatment plan, track progress, and evaluate the effectiveness of interventions. Gait analysis helped customize rehabilitation programs to address specific gait abnormalities and optimize functional outcomes for patients [10].

2.2.2 Sports and Athletics

Gait analysis finds wide application in sports and athletics, where it is employed to enhance performance and prevent injuries. Athletes benefit from a detailed analysis of their running or walking techniques, focused on factors such as stride length, cadence, posture, and foot strike patterns. By optimized these parameters, gait analysis helped to improve efficiency, identify asymmetries or inefficiencies in movement patterns, and reduce the risk of overuse injuries. Coaches and trainers utilized the insights gained from gait analysis to tailor training programs and provide targeted interventions. This lead to enhanced performance and reduced injury rates among athletes [8].

2.2.3 Biomechanics and Ergonomics

Gait assessment is a vital tool in field of biomechanics, where it contributed to the study of human movement and the forces act on the body during walking. The gait analysis is utilized to investigate kinematics (joint angles, segmental motion), kinetics (forces, torques), muscle activation patterns, and energy expenditure [8]. By quantified these parameters, biomechanics research enhanced understanding of human locomotion, provided valuable insights into normal and pathological gait patterns. Furthermore, gait analysis find applications in ergonomics, where it assessed how workplace design, footwear, and equipment impact gait patterns. This knowledge helped to optimize ergonomics to promote safety, comfort, and efficiency in various occupational and recreational settings [7].

2.2.4 Biomatric Identification

In gait analysis, concerned about safety and security are driving the adoption of advanced surveillance biometric technology. With the increased number of crimes and terror attacks, cities have integrated various biometric technologies included gait recognition into their surveillance systems. The utilization of gait recognition not only aids in identity verification but also enhances the mechanization process of surveillance recognition systems [11]. Gait is a emerged biometric modality which garnered significant attention from the research areas. Unlike modalities that require high quality images gait analysis is particularly appropriate for covert recognition in surveillance system scenarios. This is primarily due to its non-invasive nature, allow individuals to be recognized without their knowledge or interaction with the acquisition hardware [9].

2.3 GAIT ANALYSIS SYSTEM

In gait analysis, various evaluation systems are used to assess different aspects of gait identification, gait recognition, and accuracy. Gait analysis system are discussed below in detail.

2.3.1 Gait Identification

The main objectives of a gait identification is to distinguish and recognized individuals based on their unique gait patterns. This system focused on analyzing the characteristics and features of an individual's gait pattrens to establish their identification. Gait identification systems are useful in applications such as biometric authentication, surveillance, or forensic investigations. Where gait analysis served as a distinguishing factor to differentiate between individuals [12].

2.3.2 Gait Recognition/Verification

In gait recognition, individuals are recognized by their gait patterns without necessarily identified them by name or personal information. The primary purpose of this system is to classify and match gait patterns to a pre-existing database of gait templates or profiles. Gait recognition is employed in areas such as security systems, access control, or track systems, where it helped to track and monitor individuals based on their unique gait characteristics [13].

2.3.3 Accurate Evaluation

The accurate evaluation in gait analysis serves the purposed of assessing the reliability and precision of the gait identification or recognition systems. This system is designed to measure the effectiveness of the algorithms or techniques used in gait analysis. It establishes the overall precision of the system's outcomes.It involved comparing the system's output with ground truth data or known identities to evaluate the system's performance, identify potential errors or biases [14].

2.4 CATEGORIES OF GAIT ANALYSIS

In literature, there are three categories of gait analysis i.e., sensor based, sound based, and frame based gait analysis as depicted in figure 2.1.

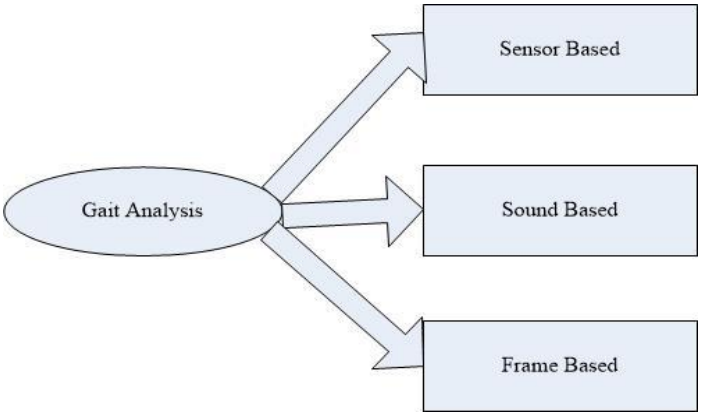


Figure 2.1: Categories of Gait Analysis

2.4.1 Sensor Based Gait Analysis

Sensor-based gait analysis employed various sensors included motion capture systems, force plates IMUs, pressure sensors, and EMG to collect data on an individual's walking pattern. Raw data is processed by filtering, noise reduction, and synchronization, leading to the calculation of critical gait parameters. These parameters are stride length, step length, gait speed, cadence, joint angles, ground reaction forces, center of pressure, and gait cycle events. These parameters are then analyzed to assess gait patterns, identify deviations from normative values and detect abnormalities. Longitudinal monitor may be employed to track changed in gait over time, particularly in rehabilitation contexts. This makes sensor-based gait analysis a versatile tool for biomechanical assessment and clinical applications. Sensors which are wear and deep learning utilized for identify intentionally generated gait changes without the need for a medical expert or gait analyzer to be present. In the study, wearable sensors were used to analyze gait [15] as depicted in figure 2.1.

Ali, B., et al. [16], presented a flexible pie-resistive pressure sensor developed using Inexpensive and swift prototyping methods. The sensor, with a grapheme Nano platelet composite is applied via spraying on a fiber-based commercial disrespectful which enabled accurate and real-time gait monitoring. It shows high sensitivity and a pressure range suitable, and when linked to a data logging and transmitting device, it differentiated different motion states and offers long-term stability. It insoles offer high linear sensitivity, approximately 0.376 kPa-1 to make them effective for captured pressure variations.

Lu, J., Y. Guo, H. Liu, and J. Gao [17], presented a technique for gait phase identification and step counted using integration of data from magneto-meters and inertial sensors. It uses gait phases by identify turning points using magnetic field amplitude. Step counting is

achieved by identifying high points or peaks in magnetic field intensity and process corresponding inertial sensor data. The proposed method shows improved resistance to interference and precise gait phase detection and step counting. The utilization of magnetometers and inertial sensors for identifying gait phases and ongoing step analysis has yielded exceptional results, with an accuracy of 96.04%.

In [18], authors explored a marker less gait analysis system using optical position and an inertial measurement unit sensor on the foot segment. While the system accurately measures temporal-spatial gait parameters, it previously had substantial errors in ankle joint angle measurements. With the inclusion of the IMU, the proposed method significantly reduces errors and shows potential as a clinical gait assessment tool for various gait conditions. The system successfully and accurately measured temporal-spatial gait parameters using data from 16 sensors. Specifically, it quantifies dorsiflexion with a precision of 3.5 ± 2.5 and plantarflexion with an accuracy of 8.7 ± 6.3.

In another study [19], the authors detected initial gait abnormalities in individuals who carry the LRRK2-G2019S mutation, linked to PD, using a an inertia-based sensor system. Participants included Patients with idiopathic PD, those with LRRK2-G2019S PD, asymptomatic carriers, and control subjects. While significant differences were noted in individuals with PD and control participants while performing different walking tasks minimal changes were detected in asymptomatic carriers. Fast walking step time correlated with Data uptake, suggested its potential as a biomarker for future studies. However, the sensor-driven examination did not reveal significant gait alterations in individuals carrying the mutation without symptoms. Minimal changed in gait patterns was observed when compare PD patients and controls across various gait tasks. Utilized sensor-based gait analysis on raw data, an accuracy of 82.5% was achieved in distinguishing these groups.

Zou, Q., et al.[34], used smartphones' inertial sensors to capture gait dynamics under unconstrained conditions. The method employs a hybrid deep neural network that combined convolutional and Recurrent Neural Networks (RNNs) for dependable gait characteristic extraction. The study finds that accelerometer data is typically superior to gyroscope data, and these two forms of data complement each other. The proposed method is practical and effective for real-world scenarios, where traditional methods may be challenging.

2.4.2 Sound Based Gait Analysis

Sound based gait analysis involved the use of audio sensors or microphones to capture sounds produced during walk. Recorded audio data is analyzed to extract key gait features like footstep timing, cadence, stride length, and footfall intensity. These parameters are then compared to normative values to assess an individual's gait pattern and identify potential abnormalities. Expert interpretation guides recommendations for interventions, such as physical therapy or gait retrained, while longitudinal monitored track change in gait over time. This method offers an alternative approach to sensor-based gait analysis, especially useful in settings where sensor placement is impractical or costly [20]. An approach that used sounds based gait analysis as illustrated in figure 2.1.

FitzPatrick, A.J., G. Rodgers.[21], explored non-invasive diagnostic techniques for early detection of implant wear and loosened in joint replacement patients. By combined acoustic emission monitor with biomechanical modeling and hip flexion angles, the study identified significant acoustic emissions. Acoustic emissions identified during the stance phase, offered insights into the underlying biomechanics. Audible squeaked during terminal stance was observed in one patient, correlated with specific hip angles and high joint loading, indicating potential wear rates. This novel approach holds promise for improved early diagnosis and patient outcomes in joint replacements. Significant acoustic emissions were noticed during the support phase of the gait cycle in patients who had undergone joint replacement surgery. These acoustic emission signals were detected within a range of 5° to 15° hip extension, highlighting the significance of sound-based analysis in the monitoring and assessment of gait

This study shows [22], acoustic gait analysis derives human gait characteristics from footstep sounds, introduced the acoustic gait. Spatio-temporal gait parameters are extracted using three signal analysis methods, using the Teager-Kaiser energy operator consistently provided the most reliable estimates. This cost-effective and non-intrusive system offers reliable estimates compared to expensive and intrusive laboratory methods like force plates and wearable sensors. Based on the acoustic gait profile and the utilization of the Teager-Kaiser energy operator, quantitative features were extracted from audio data. The statistical results indicate an HT F-Value of (13.55) with a highly significant p-value of 0.000., highlighted the relevance of these features in gait analysis. Additionally, the TKEO F-Value was 0.48 with a

p-value of 0.487, suggested a lack of statistical significance in this context. These findings were derived from the analysis of acoustic emission signals in a sound-based approach.

In [23], individuals diagnosed with acoustic neuromas encountered symptoms such as hearing loss, tinnitus, and balance problems Using three-dimensional motion analysis, gait performance was assessed in 17 patients with acoustic neuromas and 21 healthy subjects. The study identified a significant decrease in foot flexion angle during walking, suggested a potential association with vestibular disorders caused by the acoustic neuroma. These parameters are valuable in evaluated vestibular dysfunction in these patients. Three-dimensional motion analysis was employed in a study involved 38 individuals one group had an average age of 64.1±11.1 years, while the other group had an average age of 37.6±19.1 years.. The analysis was conducted using acoustic emission signals, emphasized a sound-based approach for data collection and assessment.

Ummoogum, K., D. Das, and P. Jayakumar.[24], proposed a novel machine learning-based filter to select high-quality gait audio samples for training acoustic-based gait detectors in noisy indoor environments. The filter prioritizes gait samples characterized by clear spectral attributes and minimal interference noise, result in a 25-point increase in the model's F1 score on real-world gait data. Automated the manual annotation of gait samples enhances the development of gait detection models for assessing older adults' physical and cognitive health. Gait audio samples were collected for training acoustic-based gait detectors in noisy indoor environments. A total of 6852 audio record were obtained. The achieved F1 score was 75%, and an automated machine learning (Auto ML) approach was used for this sound-based analysis.

In [25], the study shows that Individuals following unilateral hip arthroplasty exhibit impaired gait patterns even one year after surgery. Gait signification, using live feedback and instructional model sequences are enhances step length and increases hip joint range of motion during gait training. However, the impacts on step length and range of motion were not long-lasting after the cessation of acoustic feedback. Both groups showed positive changes in other gait parameters during the intervention. Real-time feedback and instructive model sequences were utilized in the context of hip arthroplasty for 22 patients. In the study, the surgical group (SG) achieved an average distance of 0.66 meters ± 0.11 meters, while the control group (CG) covered a distance of 0.60 meters ± 0.09 meters.

2.4.3 Frame Based Gait Analysis

Frame-based gait analysis employed cameras to capture an individual's walking pattern. This involved camera calibration, marker placement, and track of anatomical landmarks to calculate gait parameters. These parameters are stride length, step length, joint angles, gait speed, and cadence. Gait events are identified within the gait cycle. The data is then compared to norms to assess gait patterns and potential issues. Expert evaluation guides recommendations for interventions, like physical therapy or orthotic adjustments, while frame-based analysis also used for longitudinal monitored. This method offers detailed insights into human biomechanics and is crucial in clinical and research settings for improved gait understanding and informing treatment strategies.

In [26], the frame based technique is used to analyze the gait for different perspective it may be age detection, gender classification and move detection. Frame based are used a features are detected and apply in CNN classifier. We will examine current research models in more detail, included detailed explanations of how CNN work as shown in Figure 2.1.

Wu, Z. et al.. [27], proposed method for gait based human identification employing deep CNNs proposed method based on frame-level techniques outperformed existing methods by a substantial margin on three challenging datasets. The outcomes demonstrate that the approach is resilient to viewpoint and walking condition variations and has great potential for real-world applications. The proposed method presented a novel contribution to the literature and provides a comprehensive understand of gait-based human identification. For gait-based human identification using CNNs, outperform previous state-of-the-art methods by a significant margin on three challenging datasets. The authors demonstrate that their method is robust to viewpoint and walking condition variations and has great potential for practical applications. An extensive evaluation was conducted, and empirical results were obtained. This evaluation, using CNNs in a frame-based approach for CASIA-B dataset, yielded an accuracy of 94.50%.

The study shows in [28], identifying gait has been a demanding endeavor because of the variances in perspective, attire, and carrying circumstances. In this context, an innovative method relying on deep learning techniques auto-encoder model has been proposed to extract unchanging gait characteristics. The model generates gait features progressively by utilizing stacked multi-layer auto-encoders. It is capable of transformed transforming gait images from any perspective to a standardized side view, normalized clothing, and carrying conditions.

The suggested approach was assessed on two extensive gait datasets, and the results demonstrate its effectiveness. Gait recognition was conducted in the past to account for variations in view, clothing, and carrying conditions. This recognition The technique was trialed on the CASIA-B dataset and attained an identification rate of 89.9% using an auto-encoder in a frame-based approach.

Alotaibi, M. and A. Mahmood. [29], proposed CNN was designed frame based to be less sensitive to typical alterations and occlusions that affect gait recognition performance. It is to manage relatively compact datasets without resorting to augmentation or fine-tuning procedures. The methodology involved training the CNN on gait database, which includes multiple views and covariate aspects such as variations in carrying conditions and clothing conditions. A method was developed that was less sensitive to covariate factors such as carrying condition, clothing changes, and viewing angle variations. This approach was tested on the OUISIR-MLVP dataset and achieved an outstanding accuracy of 97.91% through a frame-based approach employing a CNN.

In [30], CNNs are used in an experimental investigation of input/output designs for cross-view gait identification. The authors suggest both unique CNN designs that are sensitive to spatial displacement brought on by subject differences as well as architectures. This is meant to be insensitive to spatial displacement brought on by variations in points of view of frame dataset. The results show that the suggested CNN architectures and score-level fusion method outperform state of the art approaches for both authentication and recognition assignments. The proposed methods are tested on two datasets with varied amounts of view differences.

For this study [31], novel deep learning approach called TGLSTM network for gait recognition. The proposed approach integrates organized data and time-related details using a frame-based learning structure, which allows the network to adaptively acquire knowledge about graphs that can evolve with time. The TGLSTM network is evaluated on four benchmark datasets based on frames for gait and action recognition. The results which show that the TGLSTM network outperforms state of the art methods on all four datasets, achieved significantly higher accuracy and lower error rates. TGLSTM network was implemented for gait and action recognition. This approach successfully addressed the problem of dynamically changing graphs. It was evaluated on raw data, achieved an accuracy of 92.7% in a frame-based manner.

Abdulhay, E et al. [32], proposed a novel approach for diagnosing Parkinson's disease using gait and tremor analysis. The methodology involved filter raw force data obtained from frame based and extracted different gait attributes using peak identification. It also represented methods for measuring pulse duration. Tremor characteristics were also analyzed using frequency dispersion and power spectrum density. For the detection of Parkinson's disease (PD) through gait analysis and recognized gait characteristics effective in distinguished between healthy individuals and those with Parkinson's disease.

A novel approach proposed for gait-based person identification using deep learning-based methods to address the issue of incomplete gait cycles was discuss in [33]. The proposed approach involved training multiple auto-encoders independently. It combined them as a uniform model to convert incomplete GEIs to complete GEIs acquired from a complete gait cycle. The proposed approach has the potential to improve the accuracy of gait recognition in real-world applications, such as video frames surveillance and security systems.

This study shows in [35], a novel approach for gait recognition across different views using MGANs and a new gait template referred to as PEI. The suggested approach surpasses several state of the art approaches on three benchmark datasets, demonstrated its effectiveness for handle variations in various angles of view and incomplete gait templates on individual frames. The use of MGANs and adversarial training allows for the learning of view-specific feature representations and the extraction of more discriminative features from gait sequences, respectively. The proposed PEI template preserves greater temporal data compared to the frequently utilized GEI template further enhanced the accuracy of the approach. These results demonstrate the effectiveness of the proposed method for cross-view gait recognition, especially when deal with variations in view angles and deficient gait templates. The use of the PEI template was found to be particularly useful for preserved temporal information and enhanced the accuracy of the approach. MGAN was developed for gait recognition, with a specific focus on address challenged related to variations in view angles and deficient gait templates. This approach was applied to the TUM-GAID dataset and achieved an impressive accuracy of 98% using MGANs in a frame-based manner.

For this [36], a convolutional neural network model based on poses that emphasizes the movement of points near human joints. The authors experimentally evaluate diverse network structures and consolidation techniques and determine that gathering supplementary data from areas surrounding the joints enhances the precision of the proposed method. The

findings illustrate the efficiency of the suggested method for gait recognition and its potential for generalization to other datasets which are based on frames. The proposed approach deviated from traditional methods that relied on full-height silhouettes of a moving person for gait recognition. Instead, this innovative approach was applied to the CASIA-B dataset it attained an accuracy of 95% using CNN in a frame-based manner.

Zou, Q., et al. [37], suggested an innovative method for gait recognition employing Convolutional Neural Networks (CNN) and optical flow derived from gait energy images. The methodology consists of two modules, motion identification and tracking, and attribute extraction, to derive gait characteristics from video sequences. The motion identification and tracking module uses a background subtraction technique to segment the mobile region associated with the spatial outline. The feature extraction module captured the gait cycle by analyzing changes in silhouette shapes, creating a short sequence of Gait Energy Images (GEI). Optical flow analysis was then employed on the GEI to isolate the moving elements, and a Convolutional Neural Network (CNN) was utilized to generate distinctive features for training and assessing the neural network. The proposed method has been expanded to encompass more extensive datasets, focusing on frames with additional human attributes. It finds utility in various applications, including healthcare monitoring, security systems, and indoor and outdoor surveillance systems. Motion detection and tracking for identification purposes, considered factors such as wearing outerwear or carrying a bag were successfully implemented. This approach yielded an accuracy rate of 97.4% using CNN on the CASIA-B dataset in a frame-based manner.

In this study [38], Gait-Net for gait recognition that integrates the steps of integrating silhouette segmentation, feature extraction, feature learning, and similarity measurement into one cohesive framework. The model consists of two Convolutional Neural Networks (CNNs) trained simultaneously with a joint loss function that combines segmentation and classification losses, and Siamese loss is added for metric learning. The experimental results demonstrate that the proposed method achieves outstanding performance on all three datasets based on individual frames. The integrated training of segmentation and recognition networks, along with the incorporation of Siamese loss, significantly streamlines the conventional stepwise process of gait recognition. This leads to significant performance improvements. An end to end network for gait recognition has been successfully developed. This network seamlessly combined multiple steps typically found in traditional gait

recognition and was evaluated using the OU-ISIR Large Population dataset, achieving an accuracy of 88.44% through a frame-based approach with CNNs.

A methodology begins with the capture of gait-related information using the slide time window method for EMG signal analysis was discuss in [39]. The CNNs are then trained using the back-propagation learning algorithm, enhanced their learning capabilities. Additionally, in the neural network architecture, a channel attention mechanism (CAM) has been incorporated to enhance the representation of gait features. The combination of CNNs and the channel attention mechanism shows promised potential for gait recognition tasks highlighted its significance in the field. There were identified issues with existing methods. These issues were addressed by implemented a Channel Attention Mechanism with CNN, resulted in an improved accuracy of 98.2% on the CASIA-B dataset in a frame-based manner.

The Gait-Net framework proposed by *Zhang, Z., et al.*[40], to gait recognition system that separates pose and appearance attributes from raw RGB frames through encoder-decoder architecture. It incorporates pose characteristics across time using a multi-layer LSTM structure. Experiments with controlled variations illustrate the efficacy of both the disentanglement approach and the integration of temporal information. The authors also provide a qualitative analysis of the disentangled features, showed their ability to capture pose and appearance information separately. It demonstrated promising computational efficiency suitable for real-time applications. Auto Encoder framework was developed for gait recognition. This framework was specifically designed to clearly separate pose and appearance characteristics from red, green and blue (RGB) images. It was tested on the OUISIR Large Population Dataset, achieved an accuracy of 94.8% in a frame-based approach.

In the study [41], gait recognition across different perspectives using deep learning techniques that addresses the problem of hard negative mining in current loss functions commonly utilized in the field of face recognition. The suggested method employs a resilient and efficient gait-related loss function known as the ACL. This approach learns multiple sub-centers for each angle within the same identity, resulting in improved intra-subject compactness. The paper also introduced a simplified spatial transformer network to identify appropriate horizontal segments of the human body and a temporal attention model based on LSTM units to learn the attention score for each frame. The suggested method has the potential to boost the efficiency of cross-view gait recognition in practical applications like

video frame surveillance and security systems. ACL function was proposed for gait recognition, and it was applied to the CASIA-B dataset. This approach achieved an impressive accuracy of 98.29% using CNN in a frame-based manner.

Wang, X. and W. Yan. [42], proposed an innovative method for gait recognition employing deep learning techniques, specifically the Conv-LSTM architecture. A novel gait representation known as ff-GEI. The authors demonstrate the effectiveness of the proposed method for cross-view gait recognition, which is important for unobtrusive video surveillance and human identification from a distance. The authors offer an extensive analysis of the experimental outcomes, encompassing confusion matrices and ROC curves, to showcase the effectiveness of the proposed approach. A technique founded on Convolutional Long Short-Term Memory (Conv-LSTM) was created to tackle the difficulties encountered in gait recognition, especially in cross-view scenarios. This method was tested on the OUISIR Large Population Dataset, achieved an accuracy of 97.91% using a CNN-LSTM approach in a frame-based manner.

In [43], proposed LFN method for gait recognition using a CNN The architecture has displayed promising results, as indicated by the experiments conducted on the OUISIR large population gait dataset. The methodology of the LFN model consists of three parallel convolutional pipelines designed to extract diverse features from the input gait silhouette image. The LFN model achieved high accuracy and stability in comparison to other gait recognition methods such as GEI and DTW. The dataset based on frames used in the experiments included normal, fast, and slow walking, as well as various clothing and carrying conditions. The resilience of the LFN model was assessed under diverse conditions, and the outcomes demonstrated that the model was robust and achieved high accuracy. Parallel convolution layers were utilized similarly to single-layer and double-layer configurations. This approach was implemented on the USF dataset, resulted in an identification rate of 98.1% using CNN in a frame-based manner.

Xu, C., et al. [44], proposed PSTN for cross-view gait identification. It included a PST and a RN, achieved state of the art performance on the publicly accessible gait datasets. The PSTN mitigates the issue of feature misalignment caused by variations in viewpoints by aligning gait features into an intermediate view through a non-rigid deformation field estimated by the PST. The registered features are then fed into the RN to output a dissimilarity score. The introduced loss function equalizes inter-subject and intra-subject fluctuations arising from

differences in perspectives, aiming to enhance recognition accuracy. The outcomes illustrate the efficacy of the suggested approach in resolving feature misalignment issues and enhancing cross view gait recognition performance on frame datasets. Architecture was specifically designed to mitigate unwanted feature misalignment caused by view differences, consequently enhanced gait recognition performance. This design was applied to the CASIA-B dataset, achieved an accuracy of 96.62% using the PSTN in a frame-based approach.

Hou, S., et al. [45], proposed SRN for gait recognition is built using SR-Blocks to concurrently extract silhouette-level and set-level features. The SR-Block incorporates a residual connection to link the two-level features, promoting increased interaction between silhouettes and sets. Additionally, it introduces an efficient approach to harness shallow features without requiring up-sampling or lateral connections. These findings underscore the effectiveness of the suggested approach in acquiring more discriminative features from the silhouettes, features that remain robust in the face of external factors like clothing, carrying conditions, and camera viewpoints, as observed in the frame dataset.

For this study [46], MvGGAN for cross-view gait recognition, that creates synthetic gait samples under various walking conditions to expand existing datasets. It enhanced the performance of deep learning-based gait classification networks. The approach comprises a generator and a discriminator, trained in an adversarial fashion to enhance the quality of generated samples. The outcomes demonstrated that the proposed method surpassed state-of-the-art techniques on all three datasets, achieving notable enhancements in accuracy. The development of MvGGAN for generated fake gait samples to extend existed gait patterns took place. This approach was applied to the CASIA-B dataset, resulted in an accuracy of 97.2% in a frame-based manner.

Sarshar, M., S. Polturi, and L. Schega [47], presented work addresses the limitations of traditional gait phase detection methods by leveraging AI, specifically LSTM algorithms, for IMU-based gait analysis. The primary objective is to enhance the accuracy and robustness of gait phase detection, particularly in situations with noisy data or individuals exhibiting variations in walking styles and physical impairments. The research introduces three supervised LSTM-based models tailored to estimate specific gait phases, namely foot-off (FOo), mid-swing (MidSw), and foot-contact (FCo). The proposed gait analysis method aims to estimate the anticipated gait phases, including FOo, MidSw, and FCo. This approach

achieved an identification rate of 97.54% using LSTM in a frame-based manner on the whuGAIT dataset.

Sepas-Moghaddam, A., et al.[48], presented method for gait recognition is grounded in deep learning and utilizes a partial feature extractor and a BGRU module. A capsule network was evaluated using two commonly used datasets with challenging test protocols. The results indicate that the proposed method surpasses state-of-the-art techniques across all four protocols, particularly excelling in demanding viewing and carrying conditions (CC). The efficacy of each component of the proposed approach was demonstrated through ablation studies. Overall, the method exhibits promising results in gait recognition and has potential applications in various domains, including video frame surveillance, forensics, and healthcare. A novel deep network was developed for gait recognition, leveraging multi-scale partial gait representations. This approach achieved a recognition rate of 96.11% on the CASIA-B dataset using Capsules in a frame-based manner

Khan, M., et al. [49], introduced a methodology for gait recognition using pre-trained CNN models on a publicly accessible gait frame dataset. They employed feature selection techniques, including Euclidean Norm, Geometric Mean Maximization, and Conditional Entropy, to extract relevant features. These selected features were aggregated using CCA for classification with various classifiers. The evaluation on the CASIA B dataset demonstrated the effectiveness of the proposed methodology in addressing the challenges associated with human gait analysis and its potential impact in the medical field. The method was also applied to detect abnormal walking patterns in suspected patients through gait analysis, achieving an accuracy of 96.58% using CNNs in a frame-based manner on the CASIA-B dataset.

Alharbi, A., et al. [50], Suggested technique validation of gait data during biomechanical rehabilitation as well as for the prediction of gait data to be utilized as a guide when constructed prostheses. In an experiment, the individual whose data were utilized to train the NN model exhibited the lowest accuracy.

In [26], model for identifying individuals using gait characteristics based on the CNN algorithm was developed. This model involved modifications to the architecture of the CNN model and the selection of hyper-parameters for various CNN model components. Additionally, this work introduced image augmentation in recognition, which enhances the

robustness of basic models against variations in images of individuals by generating images of the same frame or view under various conditions..

Bari, A. and M. Gavrilova [51], proposed a novel deep learning-based approach called KinectGaitNet was proposed for Kinect-based gait recognition. This method employed a residual learning-based convolutional neural network to process the 3D coordinates of each body joint throughout the gait cycle, eliminating the requirement for handcrafted features and avoiding gait cycle resampling by employing a global average pooling layer. The proposed method achieved state-of-the-art results on two benchmark datasets for Kinect-based gait recognition, surpassing all recently proposed state of the art methods in terms of accuracy, parameter count, and inference time. This study demonstrated the potential of the proposed deep learning architecture for real-world applications and underscored the significance of utilizing advanced techniques to enhance gait recognition performance. When applied to the KGB dataset, the proposed method utilizing a unique 3D input representation of joint coordinates achieved an impressive accuracy rate of 98.2% using Kinect-GaitNet in a frame-based manner.

Sayeed, M.S., P. Min, and M.A. Bari. [14], introduced a deep learning-based model for gait recognition, utilizing Gait Energy Images (GEI) as spatio-temporal representations. This model underwent evaluation on two publicly available gait datasets, being tested for both identification and verification tasks. For solving the verification task, a Siamese network with a parallel CNN architecture was employed. When compared to other state-of-the-art models, their proposed model exhibited higher accuracy and lower error rates. This study underscored the effectiveness of deep learning and the utilization of GEI in advancing gait recognition research. The identification and verification problems were tackled in the past using the OUISIR Large Population Dataset, achieving an accuracy rate of 93.84% through the application of a Siamese network in a frame-based approach..

The proposed framework by *Talal, E., Z. Oraibi, and A. Wali* [52], utilizes a three-part methodology involved the extraction of silhouettes, the use of CGANs to generate images, and the classification of the generated images using ResNet50 networks. The authors evaluate their framework on the training and testing set of dataset based on frames.

Sheth, A., M. Sharath, A. Reddy, and S. K. [53], objective was to create an effective gait recognition system that relies on an individual's walking pattern, using a CNN trained on a dataset. This dataset contained gait patterns, with variations in clothing and carrying items,

making it challenging for the model to accurately identify individuals. The authors achieved better results on the dataset, even with variations in clothing and baggage, which demonstrated the effectiveness of the CNN model in recognizing gait patterns under varying conditions. The remarkable accuracy of 95.45% achieved using a CNN-based approach on the CASIA-B dataset underscores the substantial potential of gait recognition as a robust biometric identification technique based on frames.

In the study [54], the regular pattern of a person's life may be compromised in some way or another by certain conditions known as neurodegenerative diseases. A prompt diagnosis is crucial for improved the patient's quality of life. Gait analysis is another tool used to analyze neurodegenerative illnesses and their stage. Use velocity-based, angle-based, sigma-lognormal features, and left walk orientation.

In [55], A novel technique called Gaussian Gait Image (GGI) is employed, which is determined across a gait cycle and represents a gait period-dependent image. The closest neighbor classifier is utilized to compute the Euclidean distance between Gaussian gait signatures based on gallery frames and probe frames. This spatial-temporal-based approach for gait recognition, utilizing GGI representations, was previously applied and achieved an impressive accuracy of 98% on the CASIA-B dataset when implemented in a frame-based manner.

Luo, J., et al. [56], introduced a gait recognition algorithm that combines two distinct features, Gait Energy Image (GEI) and Adaptive Fourier Decomposition Energy Image (AFDEI), to enhance recognition rates. Classification was performed using the nearest neighbor classifier, and the algorithm exhibited improved real-time performance. Overall, the paper presents a well-structured approach to gait recognition that offers valuable insights into the effectiveness of combining different features and its evaluation on a standard gait database based on frames. The results suggest that the proposed algorithm is suitable for real-time applications such as surveillance and security. This approach effectively preserved dynamic and static information for GEI representations and achieved a high accuracy of 98.53% on the CASIA-B dataset for gait recognition when implemented in a frame-based manner.

For this study [57], gait recognition approach proposed in this paper relies on robust feature-level fusion of four directional vectors, leveraging Gait Energy Image (GEI) and Generalized Gait Energy Image (GGEI) to extract distinctive gait features. The method is evaluated on the

largest multi-view dataset for gait recognition based on frames. The proposed approach was designed to capture various facets of the gait pattern while taking into account variations in multiple directions. When applied to the CASIA-B dataset, this approach achieved an accuracy of 98% using Support Vector Machines (SVM) in a frame-based manner.

In [39], The methodology commences with the acquisition of gait-related data using the sliding time window method for EMG (Electromyography) signal analysis. Convolutional Neural Networks (CNNs) are subsequently trained using the back-propagation learning algorithm, which enhances their learning capabilities. Furthermore, a channel attention mechanism is integrated into the neural network architecture to enhance the expression of gait features, resulting in high accuracy in gait classification. Comparative experimental results highlight the superiority of the proposed method over other CNN-based approaches. The fusion of CNNs with the channel attention mechanism demonstrates promising potential for gait recognition tasks.

Table 2.1: Review on Categories of Gait Analysis and Results

Ref.	Features	Datasets	Results	Techniques	Categorys
[52]	The proposed method focused on the side view addresses challenges in gait recognition, such as changing walking angles, wearing coats.	CASIA-B	Accuracy 95%	CGAN +resent50	Frame Based
[19]	Observed in PD patients and controlled during various gait tasks, minimal changes were detected in asymptomatic carriers	Raw Data	Accuracy 82.5%	Sensor- Gait analysis	Sensor Based
[53]	Highlighted the potential of gait recognition as a robust of biometric identification technique.	CASIA-B	Accuracy 95.45%	CNN	Frame Based
[16]	Fiber based commercial insole, enabled accurate and real-time gait monitor.	Pressure range	High linear sensitivity (~0.376 kPa-1)	Wearable electronics sensors	Sensor Based
[17]	Gait phase detection and step counted using magnetometer and inertial sensors.	Inertial sensors data	Accuracy 96.04%	Sensors	Sensor Based

[18]	System accurately measured temporal-spatial gait parameters.	16 Person sensors data	Dorsiflexion 3.5 ± 2.5 Plantarflexion8.7 ± 6.3	IMU	Sensor Based
[21]	Observation of substantial acoustic emissions during the stance phase of the gait cycle in joint replacement patients.	Acoustic	5°-15° hip extension	Acoustic emission signals	Sound based
[24]	Gait audio samples for training acoustic-based gait detectors in noisy indoor environments.	Acoustic collected recordings get 6852 audio recordings	F1=75%	Auto ML	Sound based
[25]	Real-time for feedback and instructional model sequences were employed.	Hip arthroplasty 22 patients	(SG) 0.66m ± 0.11m, (CG) 0.60m ± 0.09m	Acoustic feedback technique	Sound based
[14]	Identification and Verification Problems to solved.	OUISIR-Large Population Dataset	Accuracy 93.84%	Siamese network	Frame Based
[51]	Proposed utilized a unique 3D input represented which are coordinates of the joints. Without the need for handcrafted features or gait cycle resampling.	KGB	Accuracy 98.2%	KinectGaitNet	Frame Based
[46]	Development of (MvGGAN) for generating fake gait samples to extended existing gait patterns.	CASIA-B	Accuracy 97.2%	MvGGAN	Frame Based

[47]	Proposed gait analyses which predict the anticipated gait phases, encompassing foot-off (FOo), mid-swing (MidSw), and foot-contact (FCc).	whuGAIT	Identification rete 97.54%	LSTM	Frame Based
[48]	Development of a novel deep network for developed a novel deep network for gait recognition that leverages multi-scale partial gait representations.	CASIA-B	Recognition rete 96.11%	Capsules	Frame Based
[49]	Proposed method in detecting abnormal behavior in walking styles of suspected patients based on gait analysis.	CASIA-B	Accuracy 96.58%	CNNs	Frame Based
[39]	Issues with Existing Methods.	CASIA-B	Accuracy 98.2%	Channel Attention Mechanism with CNN	Frame Based
[43]	Parallel Convolution Layers is Used like Single layer and double layer.	USF	Identification Rate 98.1%	CNN	Frame Based
[44]	Architecture is designed to reduce unwanted feature misalignment caused by view differences, thereby improving gait recognition performance.	CASIA-B	Accuracy 96.62%	Pairwise spatial transformer network (PSTN)	Frame Based
[36]	The proposed approach departs from traditional methods that use full-height silhouettes of a moving person for gait recognition.	CASIA- B Dataset	Accuracy 95%	CNN	Frame Based
[37]	Motion Detection and Tracking based on identification of wearing the coat and holding a bag.	CASIA-B	Accuracy 97.4%	CNN	Frame Based
[41]	Proposed angle center loss (ACLo) Function for Gait Recognition.	CASIA-B	Accuracy 98.29%	CNN	Frame Based

[38]	The successful creation of a comprehensive network for gait recognition. This network integrates multiple steps involved in traditional gait recognition.	OU-ISIR Large population dataset	Accuracy 88.44%	CNNs	Frame Based
[40]	Developed a novel Auto Encoder framework for gait recognition that disentangles body orientation and visual characteristics explicitly from RGB images.	OUISIR-Large Population Dataset	Accuracy 94.8%	Auto Encoder	Frame Based
[42]	Convolutional long short term memory (Conv-L_S_T_M). Method addresses the challenges in stride identification, particularly in demanding perspective shifts.	OUISIR-Large Population Dataset	Accuracy 97.91%	CNN-LSTM	Frame Based
[31]	TGLSTM network for gait and action recognition. This approach addressed the problem of dynamically changing graphs.	Raw Data	Accuracy 92.7%	TGLSTM	Frame Based
[35]	MGAN For gait recognition specifically addressed the challenges posed by variations in view angles and deficient gait templates.	TUM-GAID	Accuracy 98%	MGANs	Frame Based
[28]	Gait recognition for variations in view, clothing, and carrying conditions.	CASIA-B	Identification Rate 89.9%	Auto-encoder	Frame Based
[29]	Presented less sensitive to consider factors like variations in load-bearing, attire alterations, and changes in viewing angle.	OUISIR-MLVP	Accuracy 97.91%	CNN	Frame Based
[57]	Considered variations in multiple directions, the proposed method captures diverse aspects of the gait pattern.	CASIA-B	Accuracy 98%	SVM	Frame Based

[27]	Extensive Evaluation and Empirical Results.	CASIA-B	Accuracy 94.50%	CNNs	Frame Based
[22]	Based on the acoustic gait profile and utilized the Teager-Kaiser energy operator.	Audio to extract quantitative	HT F-Value=13.55, P=0.000, TKEO F-Value=0.48, P=0.487	Acoustic emission signals	Sound Based
[55]	A novel spatial temporal approach for gait recognizes which utilize Gait Gaussian Image (GGIm).	CASIA-B	Accuracy 98%	GGI	Frame Based
[56]	Preserved Dynamic and Static Information For GEI Images.	CASIA-B	Accuracy 98.53%	Nearest Neighbor Classifier	Frame Based
[23]	Used three-dimensional motion analysis.	Based on 38 persons	Average age: 64.1±11.1 years old, average age: 37.6±19.1 years old	Acoustic emission signals	Sound Based

Chapter No. 03

PROPOSED METHODOLOGY

3.1 INTRODUCTION

The classification of gait as normal or abnormal movement is done using transfer learning. This approach involves collecting dataset of people walking in different styles. Datasets are pre-processed like resizing, padding and boundary extraction. Pre-trained models are used for feature extraction, train the models, fine-tuned the pre-trained models, evaluated the models performance, and deployed the models in a production environment.

3.2 METHODOLOGY

Using pre-trained models, datasets are trained on deep learning models Alexnet and AlexLSTM for gait analysis tasks. Leveraging the knowledge acquired from the pre-trained models, the transfer learning helped to enhance the model's performance and reduce the time and resources required for training. In the case of gait analysis of silhouette, AlexNet is used to extract features, train a model, and refine the pre-trained model and classify the movement of new person as normal or abnormal.

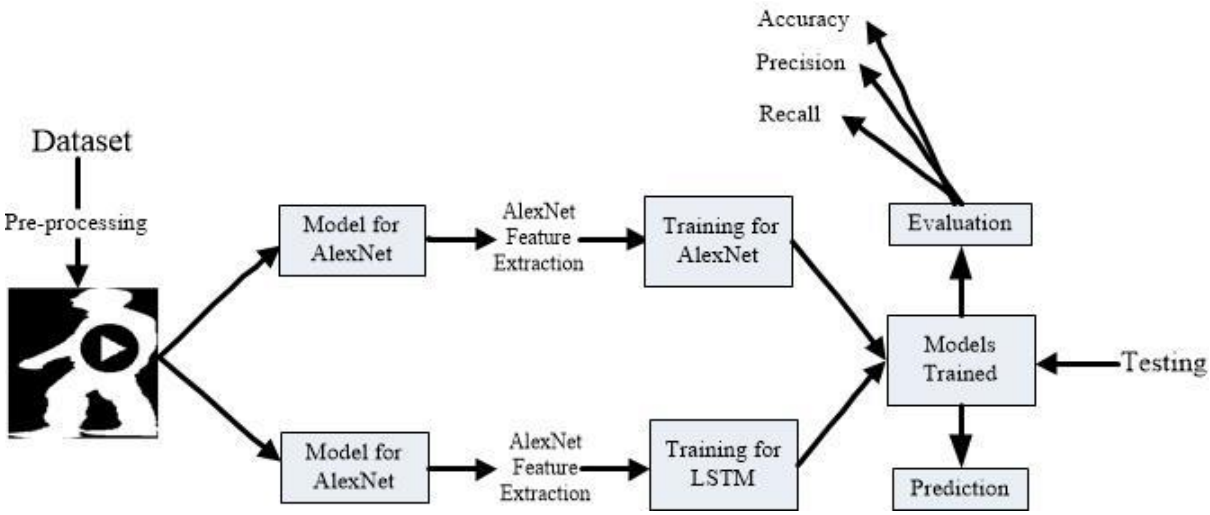


Figure 3.1: Model for Research Methodology

There are many steps to follow and accomplished the research methodology.

3.2.1 Data Collection

In first stage we have collected datasets. The dataset is important in deep learning as it provides collection of labelled data for training and validating deep learning models, specifically for the task of gait analysis. Datasets based on variety of gait patterns which

helped to improve the accuracy and robustness of the deep learning models in gait analysis. The dataset also used for transfer learning, where pre-trained models on the dataset are adapted to new gait-related task. Dataset OU-ISIR Gait Speed Transition [4], OU-ISIR dataset [58] and INIT Gait dataset [5] are used in this research.

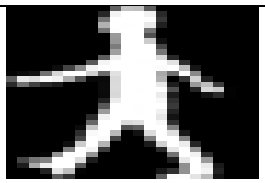
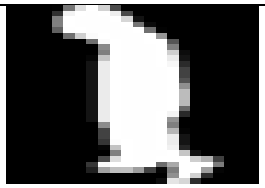
3.2.2 Datasets

3.2.2.1 OUISIR ST Dataset

The OU-ISIR Speed Transition is a dataset [4] designed to support research in the field of human gait analysis, which is focusing on gait analysis. The dataset is distributed by the Institute of Scientific and Industrial Research (ISIR) at Osaka University (OU). In OUISIR ST Dataset 2, gallery consists of gait sequences from a total of 178 subjects. These subjects were part of the data collection process. The purpose of the gallery dataset is to serve as a reference or comparison group for the analysis based on walking style. The duration of six seconds was chosen to ensure an adequate amount of gait data was captured during the steady-state walking condition and subjects are 178 and total sequences 12,180 of subjects all are resized to a standardized (22 × 32) resolution. By comparing gait patterns in different conditions like movement of subjects as head is align in proper, arm is swing and foot stride. Furthermore, the gallery dataset is used for developing and training gait recognition models. By analyzing the gait patterns of the subjects in the gallery dataset, we extract features and patterns that are used to identify individuals based on their gait characteristics.

Total of 178 subjects were assessed based on their movements, with 35 subjects classified as abnormal movement due to issues such as misaligned head, improper arm swings, and incorrect foot stride, along with other unspecified factors. The remaining 143 subjects were categorized as normal movement. For normal subject's analysis of the movement sequences resulted in 9828 and 2352 abnormal sequences. This data provides important insights into the frequency and attributes of unusual movements, potentially contributing to improved diagnosis and treatment approaches for individuals exhibiting such movement patterns. Some of examples are shown in table 3.1.

Table 3.1: View of OUISIR ST Dataset

No.	Subject Id	Frame	Status	Head Align	Arm Swing	Foot Stride
1	13		Normal Movement	✓	✓	✓
2	00222M400000		Abnormal Movement	×	×	×

3.2.2.2 OUISIR Dataset

The OU-ISIR Gait Dataset [58] is a dataset designed to support research in the field of human gait analysis, specifically focusing on age related studies such as gait-based age estimation and gait analysis of human. The dataset published by the Institute of Scientific and Industrial Research (ISIR) at Osaka University (OU).

The OU-ISIR dataset is a vast gait database with 4007 subjects (2135 males and 1872 females) aged between 1 and 94 years, records walking sequences for each subject for the Gallery modes with 6 to 15 seconds. Each subject's data consists all resized to a standardized (128 × 88) resolution.

Total of 4007 subjects were assessed based on their movements, with 515 subjects classified as abnormal movement due to issues such as misaligned head, improper arm swings, and incorrect foot stride along with other unspecified factors. The remaining 3492 subjects were categorized as normal movement. For normal subject’s analysis of the movement sequences resulted in 413521 and 63237 abnormal sequences. This data provides significant insights into the prevalence and attribute of abnormal movements, potentially contributing to improved diagnosis and treatment approaches for individuals exhibiting such movement patterns. Some of examples are shown in table 3.2.

Table 3.2: View of OUISIR Dataset

No.	Subject Id	Frame	Status	Head Align	Arm Swing	Foot Stride
1	5000021		Normal Movement	✓	✓	✓
2	0000015		Abnormal Movement	×	✓	✓

3.2.2.3 INIT GAIT Dataset

INIT Gait Dataset [5] is a publicly available dataset intended for research purposes. The videos comprising the INIT Gait Database were recorded in the Laboratory of Communication Sciences (LABCOM) at the University Jaume I. This facility is designed to support educational activities and active research in audio visual communication technologies, advertising, and journalism. The purpose of the INIT Gait Database is to serve as a benchmark for defining measures of gait regularity and validating their effectiveness in analysing abnormal gait sequences. This dataset aims to contribute in the field of clinical gait assessment, rehabilitation, medical diagnosis, and the detection of medical emergencies in hospital settings. The dataset consists of high-quality binary silhouettes extracted from videos recorded duration 10 to 17 seconds in a specialized studio within the audio-visual facilities of the University Jaume I. In total, seven different gait styles were simulated, inspired by pathological gaits associated with neurological diseases such as Parkinson's. An additional eighth style representing a normal and unaffected gait pattern was included as well. Each participant was recorded twice for each gait style, resulting in a total of 160 subjects and sequences are 41,104 of these subjects all are resized to a standardized (800 × 400) resolution.

Total of 160 subjects were assessed based on their movements, with 22 subjects classified as abnormal movement due to issues such as misaligned head, improper arm swings, and incorrect foot stride, along with other unspecified factors. The remaining 138 subjects were

categorized as normal movement. For normal subject's analysis of the movement sequences resulted in 29450 and 11654 abnormal sequences. This data provides substantial insights into the prevalence and characteristics of unusual movements, potentially contributing to improved diagnosis and treatment approaches for individuals exhibiting such movement patterns. Some of examples are shown in table 3.3.

Table 3.3: View of INIT GAIT Dataset

No.	Subject Id	Frame	Status	Head Align	Arm Swing	Foot Stride
1	108_90_a-r0_01		Normal Movement	✓	✓	✓
2	110_90_fb_01		Abnormal Movement	×	×	×

3.2.3 Pre-processing and Data Segmentation:

In order to pre-process INIT Gait dataset [5], OU-ISIR Gait Speed Transition[4] and OUISIR dataset [58] to use in deep learning model, several steps are to be taken. In data segmentation, training set of dataset is 70% and testing set is 30%.

3.2.3.1 Resize

In order to maintain consistency across all data points for AlexNet, the dataset, including INIT Gait dataset [5], OU-ISIR Gait Speed Transition [4], OUISIR dataset [58] undergoes resizing to a standard size. However, this resizing is only applied when the dimensions of the dataset are larger than the acceptable dimensions of AlexNet [59]. By following this approach, ensured that all datasets are appropriately scaled to fit the dimensions expected by AlexNet, while preserving the original size for datasets that are already compatible. We resized the dataset INIT Gait dataset from to 227x227 pixels and others remain in original size of datasets.

3.2.3.2 Padding

Padding in the pre-processing of small-size datasets involved adding extra pixels to the frames to increase their size to a desired dimension. This is useful when we want to make the images compatible with a specific input size requirement of a deep learning model like AlexNet. Small dataset OUISIR ST [4], OUISIR dataset [58] consists of silhouettes (binary images), the padding process remains essentially the same. However, we need to make sure that the padding value is appropriate for binary images for both datasets. In this case, we want to pad the images with binary values (0 or 1) to maintain the integrity of the silhouettes [59].

3.2.3.3 Bounding box Extraction

In the context of silhouette, bounding box extraction referred to the process of identifying and isolating the rectangular region that encloses the object or foreground elements within the silhouette image. Silhouette images consist of only two pixel values: typically, 0 for the background and 1 for the foreground [60]. The black portions of the silhouette are removed from the silhouette frames in OUISIR ST [4], OUISIR [58] and INIT Gait [5] datasets for gait analysis to enhance the quality and accuracy of the data.

3.2.4 Feature Extraction Using AlexNet

In order to extract features from the INIT Gait dataset [5], OU-ISIR Gait Speed Transition [4]and OUISIR LP dataset [58] which are based on human gait, AlexNet is used for the purpose of feature extraction. In a normal gait analysis situation various aspects of the body's behaviour are observed.

3.2.4.1 Normal Movement of Human

3.2.4.1.1 Head Position

The head maintained a proper position, aligned with the human body as shown in figure 3.2.

3.2.4.1.2 Arm Swing

As the gait analysis, arms swing naturally in coordination with their stride, adding balance and their movement as shown in figure 3.2.

3.2.4.1.3 Foot Stride

The feet take steps in a regular and proper manner, alternating smoothly between each step. This combination of a correctly positioned head, synchronized arm swing, and coordinated

foot movements characterizes the normal gait of a person during walking as shown in figure 3.2.

Figure 3.2: Normal Movement INIT Gait Dataset [5]

On the other hand, in an abnormal walking scenario, distinct deviations are observed.

3.2.4.2 Abnormal Movement of Human

3.2.4.2.1 Head Position

The person's head position is not aligned appropriately, possibly facing downwards instead of being upright. This improper head position affects the overall stability and balance during walking as shown in figure 3.3.

3.2.4.2.2 Arm Swing

There is an absence of proper arm swing, where the arms remain stationary or fail to move with the stride. This lack of arm movement disrupted the body's natural walk as shown in figure 3.3.

3.2.4.2.3 Foot Stride

The footsteps exhibit abnormal patterns, deviating from the regular and coordinated steps seen in a normal gait. These irregular foot movements are included dragging, shuffling, or uneven placement of the feet, leading to an unsteady and inefficient walking style as shown in figure 3.3.

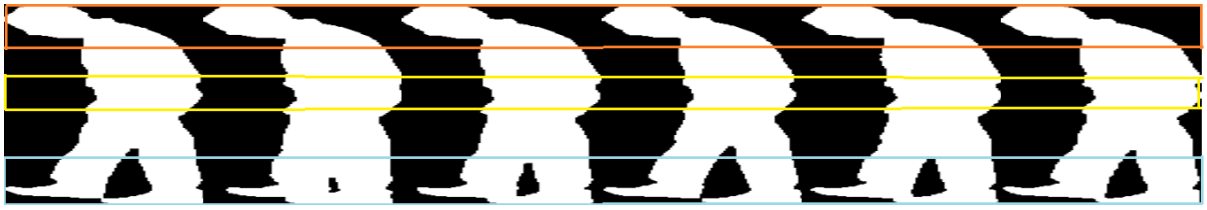


Figure 3.3: Abnormal Movement INIT Gait Dataset [5]

Video frames are separated from each video to extract the characteristics from individual silhouette frames. AlexNet, a pre-trained CNN model, is used to extract the characteristics from these frames. Pass the each frame through the model to produce a feature vector. These

feature vectors are saved with the corresponding class labels, and used as inputs to a separate deep learning model for classification.

3.3 ALEXNET AND ALEXLSTM MODELS

Models are employed for categorizing silhouettes as either normal or abnormal movement for gait analysis. In Alexnet and AlexLSTM, these approaches involved collecting datasets of silhouette gait based on frames, pre-processing these frames [49], trained the respective model on the datasets. After training models we evaluated model performance, and deployed the model in a production environment. AlexNet used as convolutional neural network to classify the frames individually from the gait sequences. AlexLSTM incorporates a Long Short-Term Memory [61] layer to better capture temporal information in the gait frames.

3.3.1 AlexNet

AlexNet is a CNN that was initially designed for image classification tasks [62]. It is used for gait classification tasks by handling the spatial features of frames. Spatial features capture spatial arrangement and relationships in data. They include shapes, edges, textures, and object arrangements. CNNs use convolutional filters to extract spatial features from input data. One approach to used AlexNet for gait classification using silhouette is to treat each frame of gait sequence, and extract silhouettes from the frames to use as input to the network [63]. AlexNet was to modify some layers for our requirement and used this model for silhouette based gait classification.

The architecture of AlexNet for gait classification using silhouette is shown in figure 3.4.

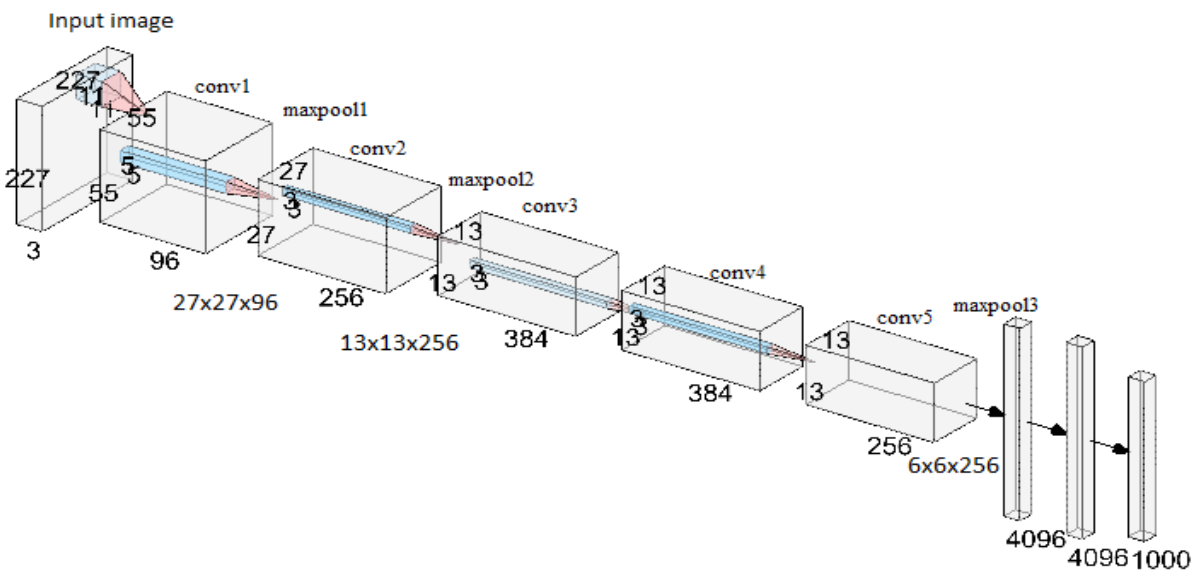


Figure 3.4: AlexNet Original Architecture

The table 3.4 provides essential information about the architecture of AlexNet, a ground breaking CNN model designed for image classification.

Table 3.4: AlexNet Architecture

AlexNet Layers	Size Of Features	Filters	Filters Size	Stride	Padding	Activation Functions
Input Layer	[227 x 227 x 3]	-	-	-	-	-
Conv1	[55 x 55 x 96]	96	[11 x 11]	4	-	(ReLU)
MaxPool1	[27 x 27 x 96]	-	[3 x 3]	2	-	-
Conv2	[27 x 27 x 256]	256	[5 x 5]	1	2	(ReLU)
MaxPool2	[13 x 13 x 256]	-	[3 x 3]	2	-	-
Conv3	[13 x 13 x 384]	384	[3 x 3]	1	1	(ReLU)
Conv4	[13 x 13 x 384]	384	[3 x 3]	1	1	(ReLU)
Conv5	[13 x 13 x 256]	256	[3 x 3]	1	1	(ReLU)
MaxPool3	[6 x 6 x 256]		[3 x 3]	2	-	-
Drop6	[6 x 6 x 256]	0.5	-	-	-	-

3.3.1.1 Layers of AlexNet

3.3.1.1. 1 Input Layer

This layer takes a size of image 227 x 227 x 3 (RGB) as input in figure 3.4. The proposed model represents the frames size of 227x 227 x 1 for silhouettes.

3.3.1.1.2 Convolutional Layer 1

As Convolutional layer [62] consists of 96 size of filter 11 x 11 x 3 with a stride of 4 for the AlexNet model in figure 3.4. The output of this layer is 55 x 55 x 96. For proposed model represents 96 size of filter 11 x 11 x 1 with a stride of 4 in figure 3.5.

$$Feature\ maps = \frac{(Input_Size - Size_of_Convolution_Kernels + 2\ Padding)}{Strides} + 1 \qquad \textbf{Equation 3.1}$$

ReLU is a non-linear activation function that is applied individually to the output of each node layer. It returns the input value if it's greater than zero otherwise it returns zero. In mathematical notation, the ReLU function is defined as follows [62].

$$F(x) = max\ (0, x) \qquad \textbf{Equation 3.2}$$

3.3.1.1.3 Max Pooling Layer 1

This layer follows the 1st convolutional layer and has a size of pool of 3 x 3 with a stride of 2. The output of this layer is 27 x 27 x 96.the whole scenarios based on Equation 3.1.

3.3.1.1.4 Convolutional Layer 2

This layer consists of 256 size of filter 5 x 5 x 48 (the input is 48 because the previous layer had 96 channels split into two) and map features with eq. (1) and padding is used 2. The output of this layer is 27 x 27 x 256.The normalization works with this second convolution layer and activation function works with Equation 3.2.

3.3.1.1.5 Max Pooling Layer 2

This layer follows the 2nd convolutional layer and has a size of pool of 3 x 3 with a stride of 2. The output of this layer is 13 x 13 x 256 and applies Equation 3.1 for features map.

3.3.1.1.6 Convolutional Layer 3

This layer consists of 384 size of filter of 3 x 3 x 256 stride 1 and padding 1. The output of this layer is 13 x 13 x 384 and applies eq. (1) for features map and use activation function ReLU with Equation 3.2.

3.3.1.1.7 Convolutional Layer 4

Layer consists of 384 size of filter of 3 x 3 x 192 (the input is 192 because the previous layer had 384 channels split into two), stride 1 and padding 1. The output of this layer is 13 x 13 x 384 and same as third convolutional layer.

3.3.1.1.8 Convolutional Layer 5

The convolutional layer consists of 256 filters of size 3 x 3 x 192 (the input is 192 because the previous layer had 384 channels split into two), stride 1 and padding 1. The output of this layer is 13 x 13 x 256 same as first and second convolutional layer.

3.3.1.1.9. Max Pooling Layer 3

The 3rd max pooling layer follows the 5th convolutional layer and has a size of pool of 3 x 3 with stride of 2. Output of this layer is 6 x 6 x 256 and applies Equation 3.1 for features mapping.

3.3.1.1.10. Fully Connected Layer1

The 1st FC layer consists of (4096) neurons with a dropout rate of 0.5.

3.3.1.1.11. Fully Connected Layer2

The 2nd FC layer consists of (4096) neurons with a dropout rate of 0.5.

3.3.1.1.12. Output Layer

The output layer [62] which uses softmax activation for layer with 1000 neurons for the AlexNet[62] in figure 3.6.

$$Pi = \frac{e^{Xi}}{\sum_j e^{Xj}} \qquad \textbf{Equation 3.3}$$

In our proposed model that represents the output layer which uses softmax activation for layer with 2 neurons which refers the 2 classes (NormalMovement or AbnormalMovement) in figure 3.5.

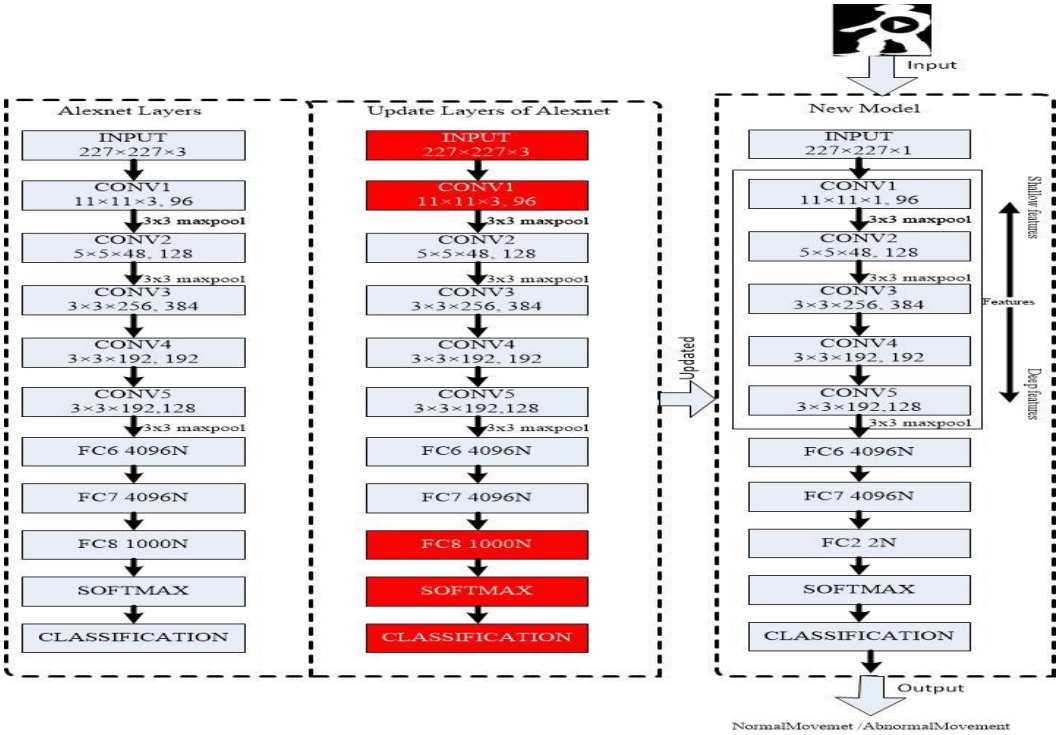


Figure 3.5: AlexNet Work Flow

3.3.2 AlexLSTM

AlexLSTM refers to a hybrid deep learning architecture that merges the capabilities of CNNs with LSTM networks for the classification tasks. The AlexLSTM is derived from the utilization of the popular CNN architecture called AlexNet in conjunction with LSTM networks [42]. The model incorporated the strengths of both architectures to effectively capture spatial and temporal features from data. Temporal features capture order and sequence in time-based data. They are important for tasks like video analysis, where understanding patterns and trends over time is crucial. The initial layers of the AlexLSTM model consist of a CNN, such as AlexNet, which processes individual frames or frames extracted from videos, enabling the extraction of meaningful visual characteristics. These characteristics are fed into the LSTM layers, which is used for capture temporal dependencies and sequence information over time.

In gait analysis for silhouette, one common approach for feature extraction is used AlexNet which is CNN based architecture. AlexNet is recognized for its ability to extract high-level features from frames effectively. However, when dealing with data, sequential information is crucial. To capture these dynamics a LSTM [61] network is used. By using AlexNet as a feature extractor, the process each frames of a video individually and extract meaningful visual silhouette features. These features are captured from various visual patterns, such as

movement of human. These extracted features serve as inputs to the LSTM network, which is a type of RNN capable of learning sequential information. It captures the sequential nature of frames and retains information over long durations. By training the LSTM network on the extracted features, it learns to comprehend the temporal patterns within the video frames and classifications based on the learned patterns.

The combination of AlexNet for feature extraction and LSTM for temporal modelling allows us to hold sequential information from data. This approach is applied to various tasks related to videos such as action identification, generating video captions, and forecasting video content. The resulting models are effectively capturing the spatio-temporal characteristics of videos and enhance performance in tasks that require understanding the dynamics of visual data over time. After feature extraction of sequence data from the video frames and put it into LSTM network. For the deeper feature are extracted from convolution layers and shown in equation 3.1 and put these sequence data feature to LSTM layer.

The working of LSTM network is discussed below in detail.

3.3.2.1 Sequence Input Layer

This layer refers to a type of data input where the elements are arranged in a specific order or sequence. Each element in the sequence is associated with a particular position or time step. The input layer takes a size of image 227 x 227 x 3. Our model represented the frames size of 227x 227 x 1 for silhouettes with number of features. Sequence data use reshapes for the generated sequence of data and transfers it into the convolutional layer.

3.3.2.2 Convolutional Layers

In AlexNet, the 1st convolutional layer employed 96 filters of size 11x11, enabling it to capture low-level features from the input image. This was followed by a second layer with 256 filters of size 5x5, which extracted higher-level features. The subsequent three convolutional layers utilized 384, 384, and 256 filters of size 3x3, respectively, further refining the learned representations. Between these convolutional layers, max-pooling operations were applied to reduce spatial dimensions [61].

3.3.2.3 LSTM Network

The Long Short-Term Memory layer is a variety of RNN layer that is commonly utilized for sequence modeling assignments, such as time series examination. It is engineered to tackle

the gradient dilemma in conventional RNNs, allowing it to grasp prolonged dependencies in sequential data.

The LSTM layer takes input sequences with a shape of (number of features), where the number of features is 9216. This number of features corresponds to the flattened output size used in AlexNet. The LSTM layer contains 2200 neurons and incorporates a dropout rate of 0.5. The number of features and neurons are not fixed and can vary depending on the particular model and task at hand. These generalized details helped us understand the overall structure of the LSTM layers and the input shape used in the AlexNet model.

3.3.2.4 LSTM Layer

The main components of an LSTM layer are as follow:

3.3.2.4.1 Input Gate

The Input Gate (it) is a component in LSTM cells of recurrent neural networks. It controlled the extent to which the fresh input should be stored in the memory cell. It combined information from the current time step with the Forget Gate and Output Gate from the previous step. This helped the LSTM handle extended dependencies and filter out irrelevant information. The gates sigmoid function and element-wise multiplication operations oversee the information flow, enabling the LSTM to efficiently manage its memory and learning processes. [64].

$$it = sigmoid(Wixt + Wihht + bi) \qquad \textbf{Equation 3.4}$$

Activation function (sigmoid)

The input gate decided how much new data from the most recent time step is introduced to the cell state. It takes input features (xt) and the previous hidden state (ht-1) as inputs and applies a sigmoid activation function to compress the values between 0 and 1. A value close to 1 indicates that the input information is important and it retained[65].

3.3.2.4.2 Forget Gate

The Forget Gate (ft) is a component in LSTM cells of recurrent neural networks. It determines what information from the preceding time step should be disregarded. The gate's sigmoid activation and element-wise multiplication control the retention of relevant information (0 for irrelevant, 1 for essential). This enables LSTMs to handle long-term

dependencies in sequential data effectively. The Forget Gate assumes a pivotal role in selective memory management for improved sequence learning [64].

$$ft = sigmoid(Wfxt + Wfht + bf) \qquad \textbf{Equation 3.5}$$

Activation function (sigmoid)

The forget gate decides which information is be discarded from the cell state. It takes input features (xt) and the previous hidden state (ht-1) as inputs and applies a sigmoid activation function. Similar to the input gate, a value close to 1 indicates that the information is retained, while a value close to 0 indicates that the information is forgotten [65].

3.3.2.4.3 Candidate Cell State

The Candidate Cell State ($\tilde{C}t$) is a component in LSTM cells of recurrent neural networks. It computed an intermediate state by combining the new input with the output from the Input Gate and the previous Cell State. This process updates the LSTM's memory cell with relevant information while preserving essential patterns from past time steps. The $\tilde{C}t$ computation helps capture extended dependencies in sequential data and enhances the LSTM's ability to manage information flow through time [66].

$$\tilde{C}t = tanh(Wcxt + Wcht + bc) \qquad \textbf{Equation 3.6}$$

Activation function (tanh)

The candidate cell state represents the new information that is potentially added to the cell state. It takes input features (xt) and the previous hidden state (ht-1) as inputs and applies the hyperbolic tangent (tanh) activation functions. The tanh function squashes the values between -1 and 1, capturing the potential new information to be added to the cell state.

3.3.2.4.4 Cell State Update

The Cell State Update is a crucial step in LSTM cells. It merges the Forget Gate, Input Gate, and Candidate Cell State to revise the extended memory (Cell State). The Forget Gate determines what information to forget, the Input Gate regulates new input, and the Candidate Cell State computes intermediate values. Element-wise multiplication and addition control the information flow. The updated Cell State retains relevant information and manages sequential data effectively [66].

$$Ct = (ft \times Ct\text{-}1 + it \times \tilde{C}t) \qquad \textbf{Equation 3.7}$$

as proven by [66].

The cell state (Ct) is updated by combining the information from the forget gate (ft), the previous cell state (Ct-1), and the input gate (it) multiplied by the candidate cell state ($\tilde{C}t$). The forget gate determines how much of the previous cell state is retained, while the input gate determines how much of the new candidate cell state is added.

3.3.2.4.5 Hidden State Calculation

The computation of the Hidden State (ht) in LSTM cells is crucial for output and short-term memory. It involves the Output Gate (ot), determining how much of the updated Cell State (Ct) to reveal. The Output Gate value is element-wise applied to Ct to produce ht. This selective exposure of information helps LSTMs capture patterns and make accurate predictions in sequential data tasks as proven by [66].

$$ht = ot \times tanh(Ct) \qquad \textbf{Equation 3.8}$$

The hidden state (ht) is calculated by applying the output gate (ot) to the updated cell state (Ct). The cell state is passed through the hyperbolic tangent (tanh) function, which squashes the values between -1 and 1, and then multiplied element-wise with the output gate. This determines how much of the updated cell state is exposed as the hidden state.

These equations outline the calculations and transformations that occur within an LSTM [61] layer, allowing it to process sequential data, capture long-term dependencies, and produce updated cell states and hidden states at each time step.

3.3.2.4.6 Output Gate

The Output Gate (ot) within LSTM cells ascertains the extent to which the updated cell state (Ct) is revealed as the hidden state (ht) or output. It employs a sigmoid function to generate values within the range of 0 to 1. An ot value of 0 hides information, while 1 fully reveals it. This selective exposure enables LSTMs to retain relevant knowledge, filter noise, and make accurate predictions in sequential data tasks. The hidden state (ht) becomes the output and input for the next time step as proven by [64].

$$ot = sigmoid(Woxt + Wohht + bo) \qquad \textbf{Equation 3.9}$$

Activation function (sigmoid)

The output gate controls how much of the cell state is exposed as the hidden state at the current time step. It takes input features (xt) and the previous hidden state (ht-1) as inputs and applies a sigmoid activation function. The output gate also determines which parts of the cell state are relevant for the current output [65].

3.3.2.5 Fully Connected Layer

The FC layer takes the learned representations from the LSTM and maps them to the desired output space. In a classification task, the fully connected layer may have neurons equal to the number of two classes NormalMovement and AbnormalMovement, and the activation function used could be softmax to obtain class probabilities.

3.3.2.6 Output Layer:

This layer employs softmax activation to generate probabilities for each class, signifying the probability of the input belonging to a specific class (NormalMovement or AbnormalMovement).

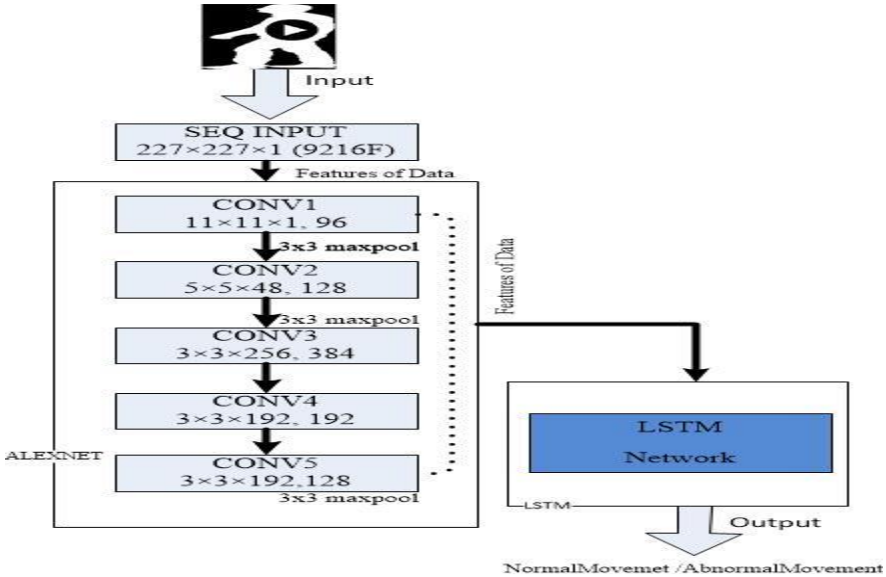


Figure 3.6: Our Work Flow AlexLSTM

3.4 MODELS GUI FOR TESTING

AlexNet and AlexLSTM are commonly used for image and video classification tasks. However, they differ in their approach and usage for frame based classification.

AlexNet was primarily designed for image classification tasks. Certainly, AlexNet comprises multiple layers followed by fully connected layers, making it suitable for image classification tasks. AlexNet can be used by treating each frame of a video as an individual image. Each

frame is fed through the network independently, and the network predicts the class label for each frame individually.

AlexLSTM, it appeared to combine the concepts of AlexNet and LSTM networks. LSTM networks are designed to model sequential data, such as time series or videos, by capturing temporal dependencies. For frame based classification, AlexLSTM is used to capture the temporal information and dependencies across frames. AlexLSTM is treating each frame independently, it processes the frames sequentially, taking into account the without temporal relationships between them.

By using AlexNet is used to processed individual frames of the video. The feature selection AlexNet's that classifies the frame into class categories normal or abnormal movement. This step treated each frame independently and performed classification based on its visual content. AlexNet based on the frame based classification, and work as shown in figure 3.5.

After obtaining frame based classifications, the feature selection from AlexNet and these features are fed into AlexLSTM. The sequence of frame based classifications serves as the input to the LSTM layers. The LSTM make predictions based on the entire frame features. This approach combines the strengths of AlexNet and LSTM which classifies the frame into class categories normal or abnormal movement as shown in figure 3.6.

We have design graphical user interface in MATLAB to facilitate the testing and evaluation of deep learning models, specifically AlexNet and AlexLSTM for the classification. The GUI provided a platform to load trained datasets for testing in a new environment. Upon loading the datasets, the GUI allowed users to apply the trained AlexNet model to classify each frame individually. If AlexNet determined that of the frames exhibited abnormal movement, it classified the person's movement as abnormal. However, the evaluation did not stop there. Next, the AlexLSTM model is classified the frames on an individual basis, analysing the sequence of frames to determine whether the movement was normal or abnormal.

GUI provided a comprehensive analysis of the silhouette as frames, incorporating the judgments of both AlexNet and AlexLSTM. It determines whether a person's movement is classified as normal or abnormal based on the evaluations of both networks and shows that the models are work correctly as shown in figure 3.7.

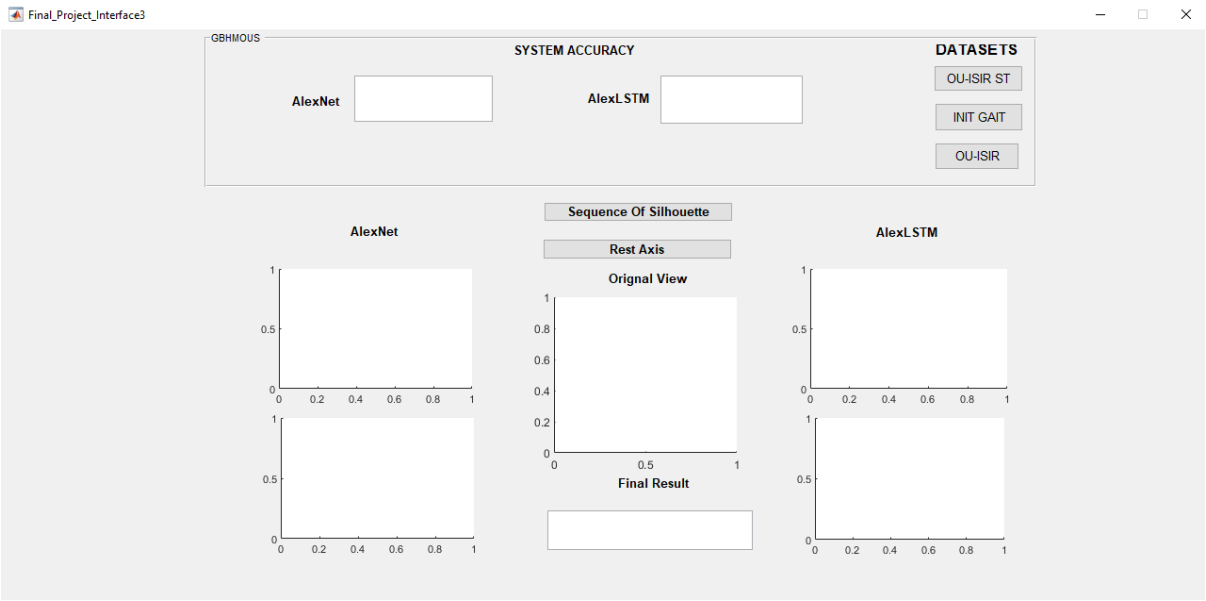


Figure 3.7: Our Graphical User Interface

Chapter No. 04

EXPERIMENTAL RESULTS AND DISCUSSION

4.1 INTRODUCTION

The experiment for deep learning models AlexNet and AlexLSTM is a fundamental stage. This chapter involved system specification and phases which are selecting appropriate models, fine-tuning experimental settings such as optimizer algorithm, learning rate and batch size. These strategies are designed to assessing model performance during both the training and testing. The models performance is compared on accuracy, recall and precision. This process helped AlexNet and AlexLSTM training and testing for datasets OUISIR ST, OUISIR and INIT Gait Dataset and produced the suitable results.

4.2 SYSTEM SPECIFICATION

System specifications using operating system Windows 10 are Intel Core i5-10500T, running at 2.30GHz with 6 cores and 12 logical processors. The system has 16GB of RAM and an 8GB GPU.

4.3 EXPERIMENTAL SETTINGS

Experimental settings are important for training deep learning models and conducted experiments. These settings determined the model learns from the data and played a significant role in affecting the model's performance. The experimental settings for the models used as transfer learning included the following:

4.3.1 Optimizer Algorithm

Adam is a popular optimization algorithm used for AlexNet and AlexLSTM models to update the model's parameters during training. Adam combined the benefits of the AdaGrad and RMSprop algorithms, making it efficient and effective for various transfer learning tasks.

4.3.2 Learning Rate

The learning rate is a hyper parameter that determines the step size at which the optimizer adjusts the AlexNet and AlexLSTM model's parameters during training. A smaller learning rate like 0.0001 means the model makes tiny adjustments to the parameters after each batch of data, which it helped to converge in better solutions, but also slow down training.

4.3.3 Batch Size

During training, the dataset is divided into smaller batches, and the model's parameters are updated based on the average gradient computed on each batch. In the OUISIR ST Dataset, when using a batch size of 2^7 AlexNet and AlexLSTM models process 128 samples simultaneously before updating their parameters. Similarly, batch sizes of 2^8 and 2^9 correspond to processing 256 and 512 samples in each batch for INIT Gait and OUISIR Datasets, respectively.

Table 4.1: Data and Parameter Settings

Dataset	**Number of Subjects**	**Number of Sequences**	**Models**	**Optimizer**	**Batch Size**	**Learning rate**
OUISIR	4007	476,758	AlexNet	Adam	(2^9)	0.0001
			AlexLSTM	Adam	(2^9)	0.0001
OUISIR-ST	178	12,180	AlexNet	Adam	(2^7)	0.0001
			AlexLSTM	Adam	(2^7)	0.0001
INIT Gait	160	41,104	AlexNet	Adam	(2^8)	0.0001
			AlexLSTM	Adam	(2^8)	0.0001

4.4 EXPERIMENT OF OUSIIR DATASET ON ALEXNET AND ALEXLSTM

In this scenario, we are evaluated the performance of two deep learning models, AlexNet and AlexLSTM on the challenging OUISIR dataset. Both models undergo training in MATLAB with varying batch sizes and are assessed over multiple epochs to understand their learning behaviour and generalization capabilities.

AlexNet started its training with an epoch of 1. At the first epoch, it achieved a training accuracy of 96.40% in approximately 04 hour 02 minutes and 28 seconds. The model demonstrated its potential to learn from the data, as indicated by a testing accuracy of 96.08%.With increased of 4, 8, 12, 16, and 20 in subsequent epochs, AlexNet consistently exhibits improved training accuracy, reaching an impressive 99.80% at the twenty epochs. However, this comes at a cost, as the training time escalates significantly to 20 hours and 29 minutes and 50 seconds for an epoch of 20 AlexNet showcases strong learning capabilities, the incremental gains in testing accuracy at 98.69%.

In contrast, AlexLSTM, a variant of the original AlexNet equipped with LSTM cells, begins its training with an epoch of 1 as well. At the first epoch, it achieves a training accuracy of 98.40% in just 02 hours 11 minutes and 26 seconds. AlexLSTM demonstrated quick learning that evidenced by a testing accuracy of 98.42%. As the increase to 4, 8, 12, 16, and 20 in subsequent epochs, AlexLSTM exhibited the training data, achieved training accuracy of 100%. However, this memorization comes with longer training times, reaching up to 14 hours and 35 minutes 01 second for epoch of 20. Despite this high training accuracy, AlexLSTM manages to maintain strong generalization with testing accuracies consistently around 99.10% to 99.14%, suggesting robustness to unseen data.

Table 4.2: Experiment of OUSIIR Dataset on AlexNet and AlexLSTM

Dataset	**Models**	**Epoch**	**Training Accuracy**	**Time Ellipse**	**Testing Accuracy**
OUISIR					
	AlexNet	**1**	96.40%	04:02:28	96.08%
		4	97.80%	11:07:57	97.47%
		8	99.20%	14:46:43	98.29%
		12	99.20%	15:46:03	98.56%
		16	99.80%	18:45:42	98.58%
		20	99.80%	20:29:50	98.69%
	AlexLSTM	**1**	98.40%	02:11:26	98.42%
		4	99.40%	07:24:41	98.96%
		8	99.60%	09:13:57	99.07%
		12	99.80%	11:06:15	99.10%
		16	100%	12:46:10	99.12%
		20	100%	14:35:01	99.14%

To compare the results of AlexNet and AlexLSTM on the OUISIR dataset, we created a bar chart that visually represents their respective training accuracies, testing accuracies, and training times for each epoch shown in figure 4.1.

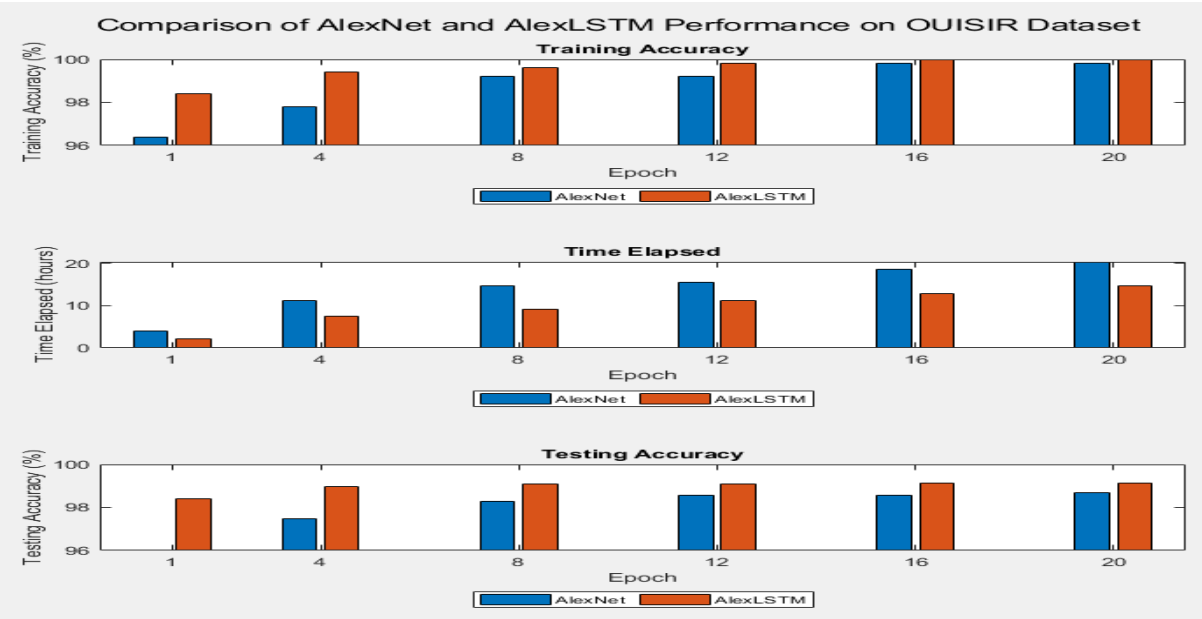


Figure 4.1: Performance Comparison of AlexNet and AlexLSTM on OUISIR Dataset

4.5 EXPERIMENT OF OUSIIR ST DATASET ON ALEXNET AND ALEXLSTM

In the evaluation of the deep learning models on the OUISIR ST Dataset, distinct learning behaviours and generalization capabilities emerged across multiple training epochs. AlexNet initiated its training with a promising start, achieving a training accuracy of 93.75% within a remarkably short time of just 6 minutes and 3 seconds in the first epoch. As training progressed through initial epoch to final epochs, AlexNet consistently improved its training accuracy, ultimately reaching a perfect 100% at epoch 20. However, this improvement came at the cost of longer training times, peaking at 39 minutes and 57 seconds for epoch 20. Intriguingly, despite achieved training accuracy, the testing accuracy plateaued at around 93.68%, suggesting that while the model learned from first epoch and training data exceptionally well, its ability to generalize for testing accuracy reach to 99.40% at twenty epochs.

AlexLSTM exhibited an entirely different learning pattern. It began its training process with remarkable speed, attaining a training accuracy of 99.22% in just 2 minutes and 2 seconds during the initial epoch. This rapid learning translated into a testing accuracy of 99.67%, indicative of strong early generalization. As training continued through multiple epochs, AlexLSTM demonstrated an extraordinary ability to memorize the training data, consistently achieving a perfect training accuracy of 100% at 20 epochs. Despite this memorization, AlexLSTM maintained robust generalization, with testing accuracies consistently hovering

around 99.81%. Furthermore, despite the increase in epochs, the training times for AlexLSTM remained reasonable, peaking at 9 minutes and 15 seconds for epoch 20.

AlexNet and AlexLSTM models exhibited impressive learning capabilities the results suggest that AlexLSTM may be the preferred choice for this dataset due to its rapid learning, strong generalization, and reasonable training times.

Table 4.3: Experiment of OUSIIR ST Dataset on AlexNet and AlexLSTM

Dataset	**Models**	**Epoch**	**Training Accuracy**	**Time Ellipse**	**Testing Accuracy**
OUISIR ST					
	AlexNet	**1**	93.75%	00:06:03	93.68%
		4	98.44%	00:14:24	98.41%
		8	100%	00:35:00	98.80%
		12	100%	00:36:20	98.99%
		16	100%	00:37:49	99.32%
		20	100%	00:39:57	99.40%
	AlexLSTM	**1**	99.22%	00:02:02	99.67%
		4	100%	00:06:09	99.77%
		8	100%	00:07:59	99.79%
		12	100%	00:08:06	99.79%
		16	100%	00:08:21	99.81%
		20	100%	00:09:15	99.81%

To facilitate a comparative analysis between AlexNet and AlexLSTM on the OUISIR ST Dataset we create a bar chart presenting their training accuracies, testing accuracies, and training times across different epochs. After training both models separately, record their training accuracies at the end of each epoch and evaluate them on the testing dataset. Additionally, measure the time each model takes to complete each epoch during training. Organize this data and create a bar chart with epochs. Analyze the chart to draw conclusions about the relative performance of AlexNet and AlexLSTM as shown in figure 4.2

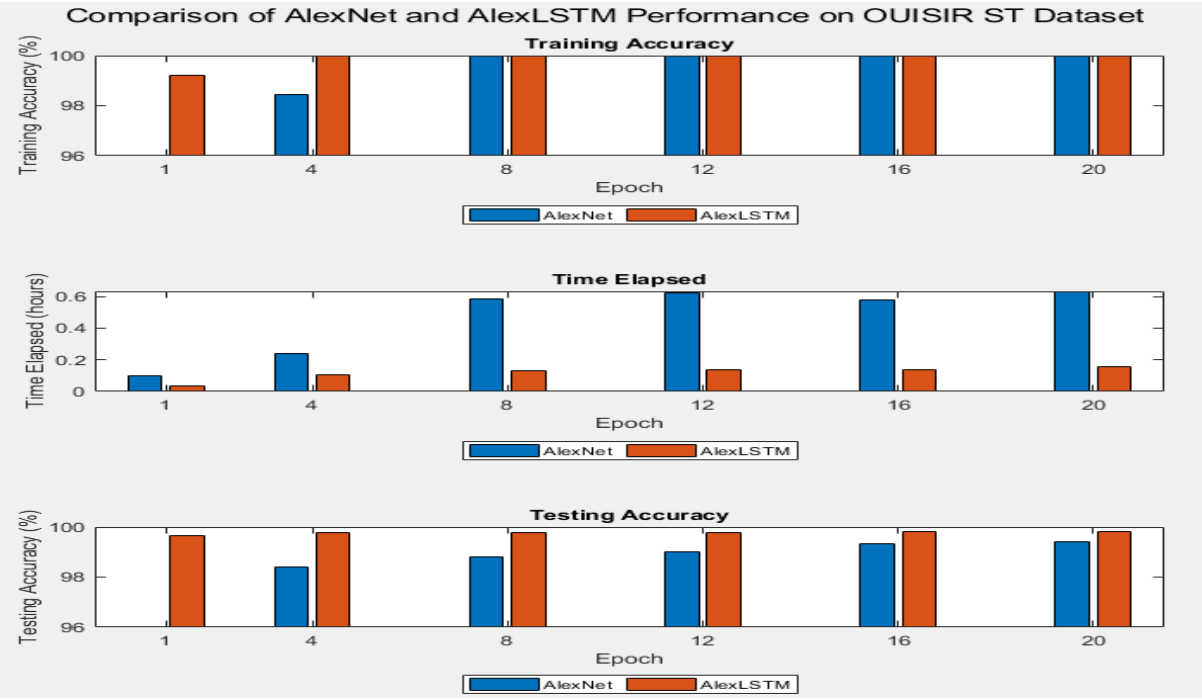


Figure 4.2: Performance Comparison of AlexNet and AlexLSTM on OUISIR ST Dataset

4.6 EXPERIMENT OF INIT GAIT DATASET ON ALEXNET AND ALEXLSTM

We conducted a comprehensive evaluation of two deep learning models, namely AlexNet and AlexLSTM, utilizing the INIT Gait dataset. Our aim was to assess their performance, learning dynamics, and generalization capabilities across a range of training epochs. AlexNet, CNN initiated its training from first epoch. Impressively, during the inaugural epoch it swiftly achieved a training accuracy of 99.20%, all within a reasonable training duration of 17 minutes and 15 seconds. The model exhibited early indications of effective learning, accurately predicting 99.25% of the test data. With an increase in the number of epochs to 4, AlexNet displayed significant enhancements in training accuracy, ultimately reaching an impressive 99.60%. However, these notable improvements came at the expense of considerably prolonged training times, approximately 54 minutes and 40 seconds. Correspondingly, the testing accuracy followed an upward trajectory, peaking at 99.55%, corroborating the model's proficiency in generalization. As training continued through epochs 8 to 20, AlexNet eventually achieved training accuracy of 100% by the 20th epoch, underscoring its capability to memorize the training data. Nevertheless, with the escalation of epochs, training times surged significantly, reaching a peak of 1 hour and 36 minutes and 43 seconds. It's remarkable that even though the model achieved outstanding training accuracy, the testing accuracy lingered around 99.71%, signalling a possible constraint in generalization beyond a specific epoch.

AlexLSTM embarked on its training starting from epoch 1. Remarkably, within 4 minutes and 1 second AlexLSTM accomplished a training accuracy of 100%. This rapid learning translated into an outstanding testing accuracy of 99.94%. As the number of epochs progressed to 4, 8, 12, 16, and 20 it consistently showcased a remarkable aptitude for memorizing the training data, consistently attain training accuracy of 100%. Notably, despite this capacity for memorization AlexLSTM adeptly maintained exceptional generalization with testing accuracies consistently hovering around 99.96%. However, AlexLSTM as we extended the number of epochs, training times also witnessed an increase, reaching a pinnacle of 43 minutes and 2 seconds for the 20th epoch. The performance of AlexLSTM underscored its robustness as the testing accuracy remained consistently high, affirming an admirable equilibrium between learning and generalization.

Table 4.4: Experiment of INIT Gait Dataset on AlexNet and AlexLSTM

Dataset	**Models**	**Epoch**	**Training Accuracy**	**Time Ellipse**	**Testing Accuracy**
INIT Gait					
	AlexNet	**1**	99.20%	00:17:15	99.25%
		4	99.60%	00:54:40	99.55%
		8	100%	01:24:15	99.59%
		12	100%	01:29:47	99.62%
		16	100%	01:30:38	99.62%
		20	100%	01:36:43	99.71%
	AlexLSTM	**1**	100%	00:04:01	99. 94%
		4	100%	00:12:30	99.94%
		8	100%	00:16:54	99.95%
		12	100%	00:18:16	99.95%
		16	100%	00:40:06	99.96%
		20	100%	00:43:02	99.96%

To conduct a side-by-side evaluation of AlexNet and AlexLSTM on the INIT Gait dataset, we've designed a bar chart that offers a visual depiction of their training accuracies, testing accuracies, and training durations for every epoch as shown in figure 4.3.

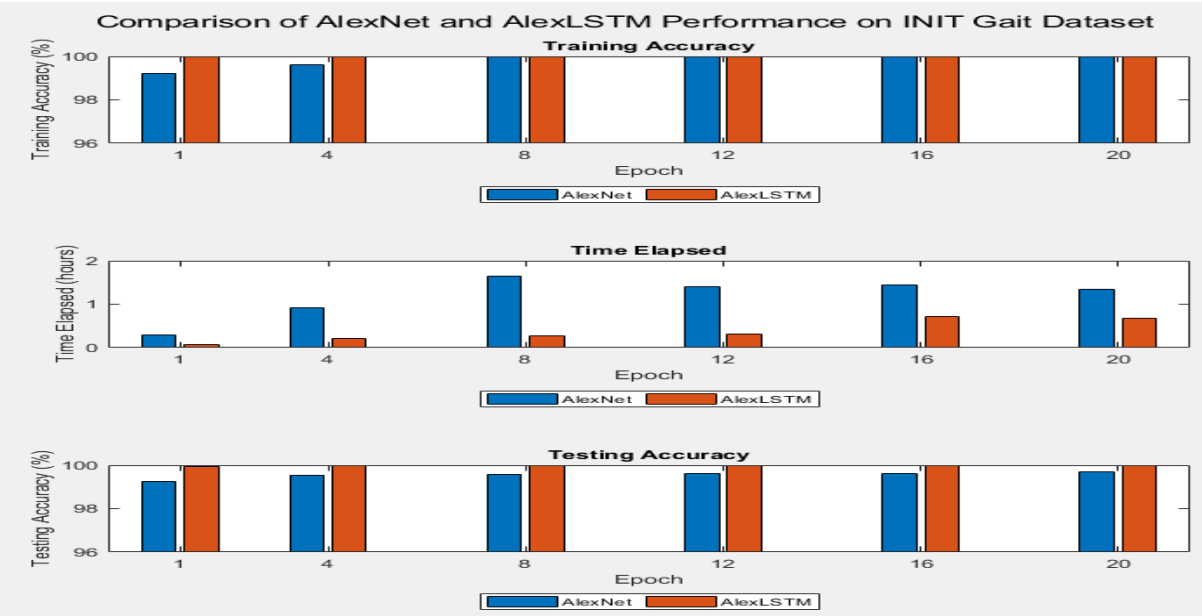


Figure 4.3: Performance Comparisons of AlexNet and AlexLSTM on INIT Gait Dataset

4.7 ALEXNET AND ALEXLSTM TRAINING AND RESULTS

Monitoring and understanding the training progress in MATLAB involves addressing various challenges such as over-fitting, resource constraints, hyper-parameter tuning, data quality, and convergence issues. Careful consideration of these limitations is essential to ensure effective and efficient model training.

During the training process of a model using a datasets in MATLAB, significant progress was observed as the model learned and adapted to the underlying patterns within the data. As the iterations and epochs of training continue, the model gradually improves its ability to make accurate predictions or classifications, leading to reduced training loss and enhanced performance on test data.

4.7.1 Alexnet Training and Results

The performance of the AlexNet and its variants was evaluated over multiple training epochs. The baseline AlexNet demonstrated strong learning capabilities on the OUSIIR Dataset, with its training accuracy improving from 96.40% to 99.8% after 20 epochs. However, it came at the cost of significantly increased training time, which escalated from 1 epoch to 20 and taking more than 84 hours for whole process. The testing accuracy eventually plateaued at 98.69% after 20 epochs as shown in figure 4.4.

In contrast, the variant of AlexNet designed for the OUSIIR ST Dataset exhibited a promising start, achieving a training accuracy of 93.75% in just one epoch. It continued to

improve and reached 100% training accuracy over five epochs, although the training time also increased, going from 1 epoch to 20 take more than 2 hours. The testing accuracy stabilized at a high level of 99.40%.

When applied to the INIT Gait Dataset, the standard AlexNet displayed rapid early learning, achieving a training accuracy of 99.22% in just one epoch. It continued to improve and reached 100% training accuracy in only three epochs. However, the training time for the1 to 20 epochs was more than 7 hours. The testing accuracy remained consistently high, hovering around 99.71%.The training dataset OUISIR on AlexNet showed the training progress in figure 4.4

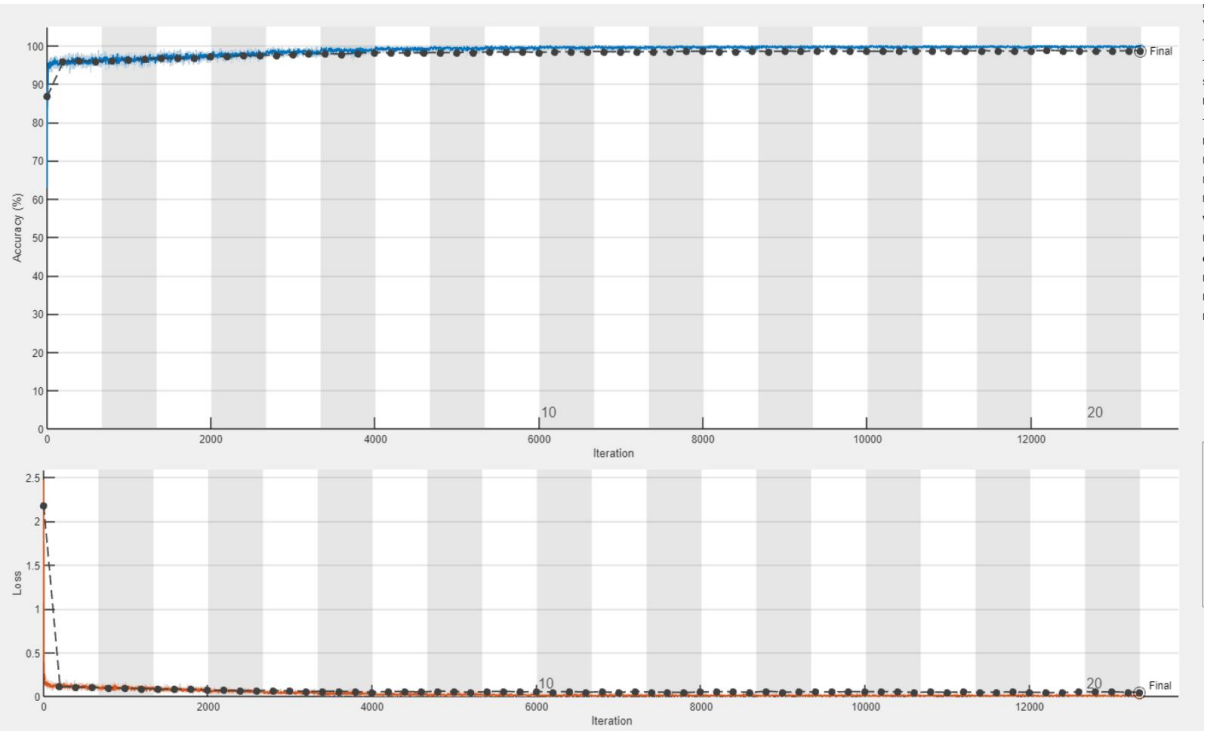


Figure 4.4: Training Progress of AlexNet on OUISIR Dataset

4.7.2 AlexLSTM Training and Results

AlexLSTM, a variant of AlexNet enhanced with LSTM cells, starts with epoch 1. On OUSIR Dataset it rapidly achieved high training accuracy of 98.40% after first epoch, with strong generalization to 98.42% testing accuracy. Over subsequent epochs, it perfectly memorizes training data with 100% accuracy, yet this extends training times up to 49 hours from first epoch to twenty. Despite perfect training accuracy, it maintains robust generalization at 99.14% testing accuracies. On OUISIR ST Dataset learns with 99.22% training accuracy in first epoch, translating to 99.67% testing accuracy. As epochs increase, it consistently

achieves perfect 100% training accuracy, with longer training times. Its generalization remains strong, consistently reaching around 99.81%. On INIT Gait Dataset with first epoch of AlexLSTM demonstrated rapid learning, hitting 100% training accuracy and have the generalization at 99.94% testing accuracy. Across epochs, it consistently achieves perfect 100% training accuracy. Despite training times more than two hours, it exhibits exceptional generalization at 99.96% testing accuracy showcasing a balanced learning-generalization trade-off. The training dataset OUISIR on AlexLSTM showed the training progress in figure 4.5

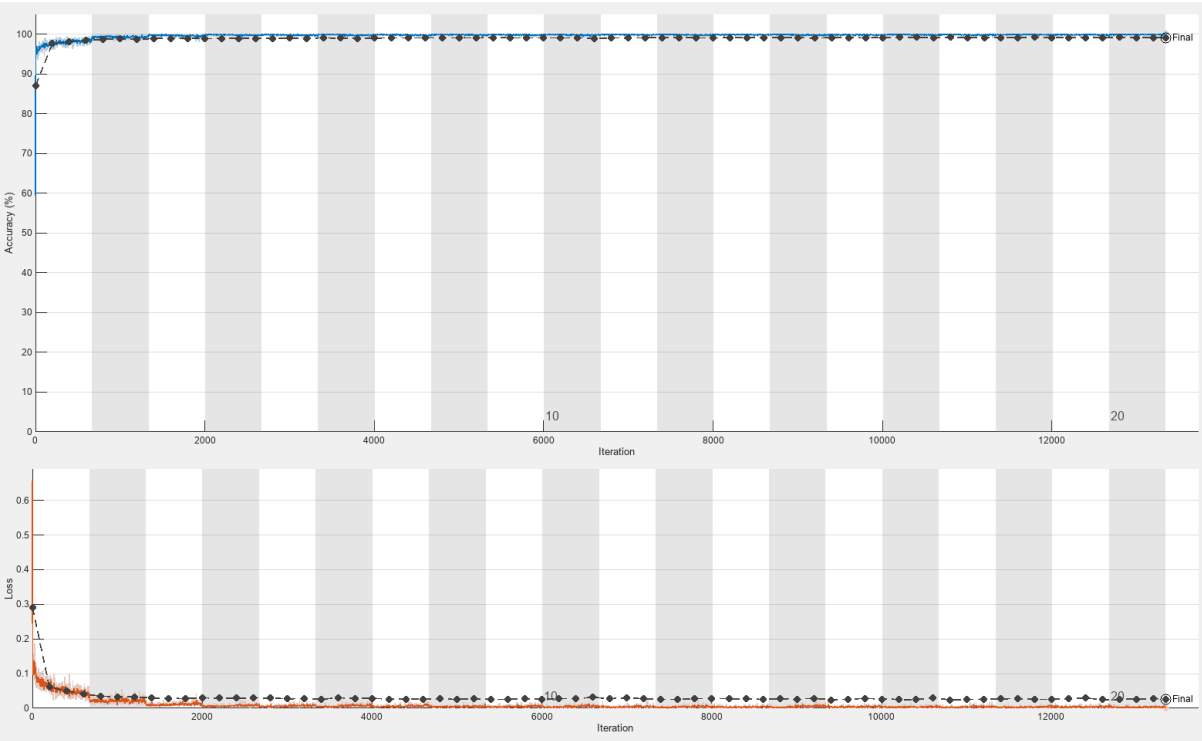


Figure 4.5: Training Progress of AlexLSTM on OUISIR Dataset

4.8 MODELS EVALUATION AND COMPARISON

A classification model's effectiveness is evaluated using three key metrics accuracy, precision, and recall. The number of accurately predicted values in proportion to all anticipated values may be determined using accuracy, but it's vital to realize that additional metrics, like as recall and precision, must also be used.

4.8.1 Recall

Recall also known as sensitivity or true positive rate measures the ability of the model to correctly identify positive instances. Recall calculates the proportion of correctly predicted

positive instances (true positives) out of all actual positive instances (true positives and false negatives). Recall provides insights into the model's ability to capture all positive instances in the dataset. It is particularly important when the cost of false negatives is high and we want to minimize incorrect negative predictions[67].

$$Recall = \frac{TP}{TP+FN} \qquad \textbf{Equation 4.1}$$

4.8.2 Precision

Precision focuses on the correctness of positive predictions made by the model. Precision calculates the proportion of correctly predicted positive instances (true positives) out of all instances predicted as positive (true positives and false positives). Precision indicates how well the model correctly identifies positive instances out of all instances that it predicted as positive. It is particularly useful in cases where the cost of false positives is high and we want to minimize incorrect positive predictions [67].

$$Precision = \frac{TP}{TP+FP} \qquad \textbf{Equation 4.2}$$

4.8.3 Accuracy

Accuracy measures the overall correctness of the predictions made by a model. It is calculated as the ratio of the correctly predicted instances to the total number of instances in the dataset. Accuracy gives an overall assessment of how well the model performs across all classes. However, it can be misleading if the dataset is imbalanced, meaning that the number of instances in different classes is significantly different[67].

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \qquad \textbf{Equation 4.3}$$

The provided data showcases the performance evaluation of models on three distinct datasets OUSIIR, OUISIR ST, and INIT Gait. Each dataset was tested using two different models AlexNet and AlexLSTM. The evaluation metrics considered include accuracy, precision, and recall. First, the OUSIIR dataset as AlexNet achieved an impressive accuracy of 98.69%, with precision and recall scores of 92.36% and 97.66% respectively. The AlexLSTM model exhibited even higher performance, attaining an accuracy of 99.14%, along with precision and recall scores of 95.14% and 98.21% respectively. Secondly, OUISIR ST Dataset the results were equally remarkable. AlexNet demonstrated a remarkable accuracy of 99.40%, and its precision and recall were 98.58% and 98.31% respectively. The AlexLSTM model

excelled even further, achieving an astonishing accuracy of 99.81%, while maintaining exceptional precision and recall values of 99.72% and 99.29% respectively. Lastly, the INIT Gait dataset produced exceptional outcomes as well. AlexNet showcased remarkable accuracy, reaching an impressive 99.71%, accompanied by precision and recall scores of 99.08% and 99.88%. The AlexLSTM model's performance on this dataset was truly outstanding, with a near-perfect accuracy of 99.96%, alongside precision and recall scores of 99.94% and 99.91% respectively.

These findings underscore the capabilities of both AlexNet and AlexLSTM models in accurately classifying and predicting outcomes across diverse datasets. Their consistently high accuracy rates, complemented by robust precision and recall metrics, validate their effectiveness in a range of applications, particularly those related to gait analysis.

Table 4.5: Model Evaluation on Datasets

S.no	Dataset	Models	Accuracy	Precision	Recall
1.	**OUSIIR**	AlexNet	98.69%	92.36%	97.66%
		AlexLSTM	99.14%	95.14%	98.21%
2.	**OUISIR ST**	AlexNet	99.40%	98.58%	98.31%
		AlexLSTM	99.81%	99.72%	99.29%
3.	**INIT Gait**	AlexNet	99.71%	99.08%	99.88%
		AlexLSTM	99.96%	99.94%	99.91%

The provided data is visually represented through a graphical diagram to illustrate the assessment of the performance for different models on the diverse datasets. The graph would have three datasets (OUSIIR, OUISIR ST, and INIT Gait) and it represent the different evaluation metrics (Accuracy, Precision, and Recall) as in percentages. Each dataset bars, one for the performance of AlexNet and the other for AlexLSTM, allowing for a clear comparison between the two models as shown in figure 4.6.

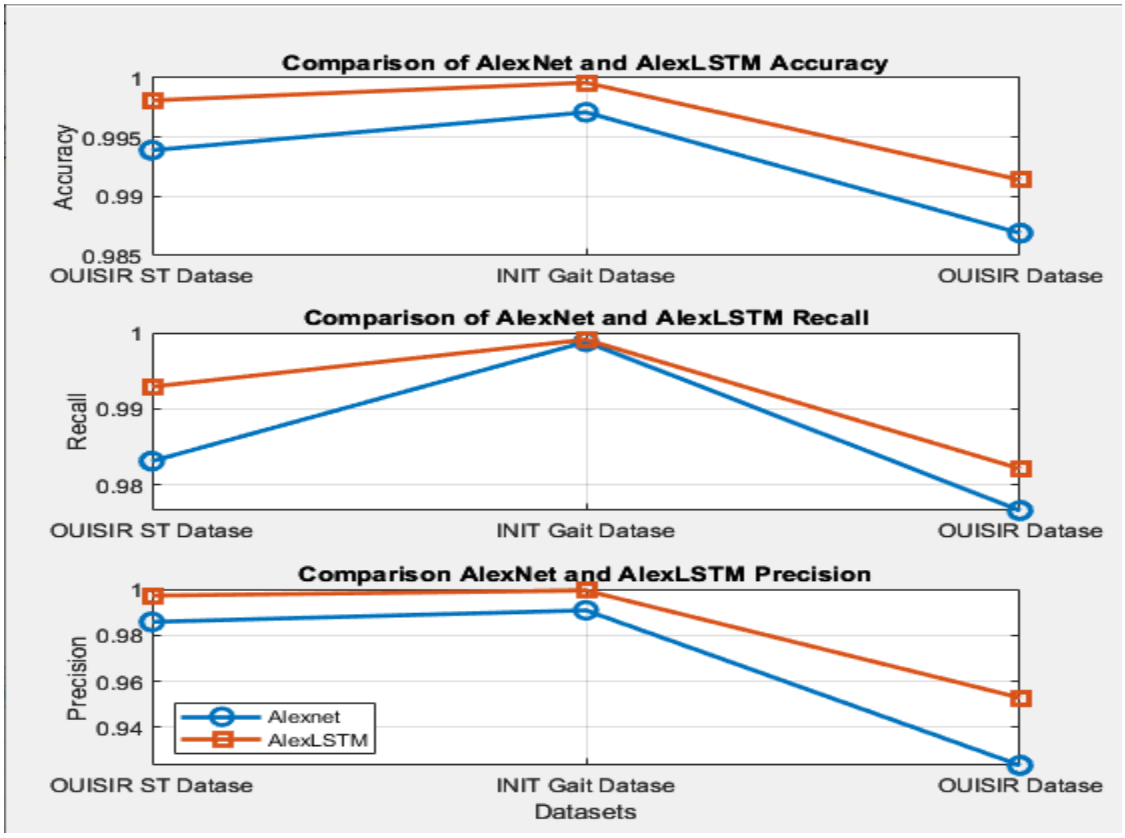


Figure 4.6: Comparison of Accuracy, Precision and Recall on Datasets

4.9 SUMMARY

In the experimental phase for evaluating the deep learning models, namely AlexNet and AlexLSTM, constitutes a pivotal aspect of this study. This chapter encompasses the systematic specification of the experimental setup, involving critical phases such as the selection of appropriate models and the fine-tuning of essential experimental settings, which include the optimizer algorithm, learning rate, and batch size. The overarching purpose of these strategies is to meticulously assess the performance of these models during both the training and testing phases. The metrics used for evaluating model performance primarily encompass accuracy, recall, and precision. This rigorous and systematic process guided the training and testing of AlexNet and AlexLSTM using three distinct datasets, namely OUISIR ST, OUISIR, and INIT Gait Dataset, ultimately yielding promising and suitable results.

CONCLUSION AND FUTURE WORK

The analysis of human movement is specifically how individuals walk and run is of great significance in various societal areas, including sports, healthcare, and military recruitment. Movement observation is a structured process aimed at gathering precise information about how individuals move, serving various goals in field such as in military. This practice has significant impacts on clinical and environment care. Gait analysis, involving the evaluation of biomechanical and kinematic factors, serves as a valuable tool for diagnosing gait disorders and can be categorized into gait identification and verification. A gait recognition method is utilizing sensor-based, sound-based, and frame-based approaches to understand and differentiate individuals based on their distinct movement patterns. In the realm of military fitness assessment, the application of transfer learning techniques using neural network architectures like AlexNet and AlexLSTM has produced impressive results. On silhouette based gait datasets (OUISIR, OUISIR ST, and INIT) the accuracy of AlexNet and AlexLSTM was 98.69%, 99.40%, 99.71%, 99.14%, 99.81%, 99.96% respectively. This innovative approach holds promise for applications in both medical and military fields, offering the potential to enhance the assessment of military personnel's fitness levels during recruitment and throughout their service. It contributes to improved gait recognition and its multifaceted implications.

In our future endeavours, we recognize a pressing need to address the absence of a comprehensive dataset tailored for gait analysis concerning military personnel, encompassing both normal and abnormal gaits. Our forthcoming work will concentrate on meticulously gathering and curating a dataset sourced from military recruits and training sessions. This dataset will encompass a diverse array of walking styles exhibited by individuals within the military context. By constructing a specialized gait-based dataset, we aim to cater to the unique movement patterns of military personnel, enabling precise analysis of their gait characteristics.

Furthermore, our research will extend to the development of novel methodologies, capitalizing on this newly created military-specific dataset. We intend to evaluate the performance of these methodologies in comparison to both existing approaches and our established methodology. The ultimate objective is to enhance the accuracy and efficacy of gait analysis for military personnel. Crucially, our future work will not remain confined to the laboratory; rather, we aspire to deploy our refined methodologies in the dynamic environment

of military recruitment. By integrating these advancements into the realm of movement observation within military contexts, we aim to provide valuable insights and improved evaluation techniques for the assessment of recruits' movement patterns. This holistic approach promises to address the specific challenges faced by military personnel and contribute to the refinement of movement analysis protocols, thereby advancing the field's practical applications.

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